Too Far to Matter? The Urban Reach of Rural Territorial Conflict

Sebastian Ritter
AQR-IREA
University of Barcelona

November 2, 2026

Latest version

Can persistent territorial conflict affect nearby urban economies even when conflict events occur outside the city? I study this question in southern Chile, where the Mapuche self-determination conflict is recurrent and predominantly rural. I link geocoded attacks, coercive actions, and land invasions to a panel of 104 fixed urban areas and measure annual external exposure from events occurring within 25 kilometers but outside the receiving area. Greater external conflict exposure is associated with fewer registered enterprises and lower tax-reported firm sales. Within an urban area, an increase from the 25th to the 75th percentile of positive annual exposure is associated with about 2.2 percent fewer enterprises and 5.3 percent lower sales. Administrative registration activity has a negative full-period association, while recorded terminations show no clear association in 2005–2015, before an administrative break. These administrative series do not identify how enterprise stocks adjust. Complementary measures of nighttime lights and newly built area show no comparable contemporaneous decline. Fiscal appraisal also does not decline contemporaneously, but turns negative at medium lags. The formal-activity association is not systematically stronger where historical Títulos de Merced cover more surrounding land or where the baseline Mapuche population share is higher. The results show that the economic reach of recurrent territorial conflict can extend beyond contested land and into nearby urban economies, even in a setting where property, tax, Indigenous-land protection, and restitution institutions continue to operate.

Keywords: Indigenous conflict; political violence; local economic activity; spatial spillovers; Chile.
JEL codes: D74; O18; R11.

1 Introduction

Conflict is usually studied where violence takes place, but its economic consequences need not stop at the location of the event. Firms in nearby towns may respond when recurrent conflict changes perceived security, access to customers and suppliers, or expectations about the future use of surrounding land. This paper asks whether persistent territorial conflict reaches nearby urban economies even when the conflict itself occurs outside them.

A large literature documents the economic consequences of conflict for directly exposed places and firms. Armed conflict reduces firm sales and production inputs, increases plant exit, and can alter employment and investment, while violence and insecurity are capitalized into property and other assets (Camacho and Rodriguez, 2013; Del Prete et al., 2023; Besley and Mueller, 2012; Brancati et al., 2026). Related evidence from Afghanistan and Mexico shows that the response of firms also depends on how violence affects security, state capacity, and local labor markets (Neri-Lainé, 2026; Utar, 2025). Much of this evidence, however, concerns firms located where conflict or insecurity occurs. A smaller literature studies how violence propagates across space. Alfano and Cornelissen (2026) show that attacks along transport routes in Somalia affect food prices in distant destination markets, while Liu et al. (2026) emphasize how trade-network connectivity conditions the transmission of violence across Nigerian markets. These studies show that the economic geography of conflict can be broader than the geography of violence itself, but they observe specific trade connections through which the shock propagates. Less is known about whether recurrent conflict around a territory affects nearby urban economic activity.

Southern Chile provides a distinct setting in which to study this question. The Mapuche conflict is an Indigenous self-determination conflict centered on territorial claims, land restitution, and political recognition rather than a conventional civil war (Cayul et al., 2022). Its geography reflects the military incorporation of Mapuche territory into the Chilean State in the late nineteenth century, the creation and subsequent fragmentation of *Títulos de Merced*, the reversal of twentieth-century land transfers, and the persistence of claims over land later transferred to non-Mapuche owners (Jaimovich and Jordán, 2025b,a; Jaimovich and Toledo-Concha, 2026). Contemporary attacks and land invasions are therefore rooted in a historically structured geography of territorial claims. Of the 2,775 eligible conflict events, 2,288 (82.5 percent) occur outside all study urban areas.

The institutional environment is also unusual relative to many settings in the conflict literature. Contemporary territorial conflict does not take place in an institutional vacuum. The 1993 Indigenous Law regulates Indigenous land transfers and created a public registry through which Indigenous-land status is enforced; CONADI operates a formal land fund through which the State purchases and restores land associated with recognized claims (Gobierno de Chile, 1993; Jordán et al., 2025; Jaimovich et al., 2025). Registry protection has slowed the loss of Indigenous ownership, while restitution has changed some land uses and continued during the conflict (Jordán et al., 2025; Jaimovich et al., 2025). Related work from other Indigenous settings shows why this distinction matters. Legal recognition and property institutions can alter bargaining power and economic outcomes (Feir et al., 2023). Albertus (2026) shows that recognition during Peru's civil conflict reduced violence partly by moving land disputes into state institutions. Southern Chile therefore makes it possible to study whether recurrent territorial conflict has economic costs beyond contested land even when the State continues to register property, collect taxes, regulate Indigenous land, and operate a formal restitution process.

This paper also differs from the existing evidence on the economic cost of the Mapuche conflict.

Larroulet Vignau and Sepúlveda (2025) estimate aggregate losses in regional GDP associated with violence in Araucanía. I instead exploit repeated variation in the location and intensity of conflict across 104 fixed urban areas in Biobío, Araucanía, Los Ríos, and Los Lagos. This four-region corridor is selected because historical Mapuche territorial occupation, recognized claims, contemporary restitution, and recent conflict overlap there (Comisión Verdad Histórica y Nuevo Trato con los Pueblos Indígenas, 2008; Jaimovich and Jordán, 2025b; Jaimovich et al., 2025; Cayul et al., 2022). Appendix Figure A.1 shows the corridor alongside a historical map of Mapuche territory and the 1992 Mapuche population share. Each area is constructed from the 2017 INE urban census-block geography and held fixed over time; these are geographic areas, not functional labor-market areas. The unit of interest is the receiving urban area rather than the region where the conflict occurs. This makes it possible to separate conflict taking place inside an urban area from conflict occurring nearby, and to ask directly how far the association with economic activity extends. I also compare formal business outcomes with nighttime lights, fiscal appraisal, and newly built area. This distinction helps establish whether the economic reach of nearby conflict appears mainly in administrative records of formal activity or is accompanied by broader changes in the physical development of cities.

I link these fixed urban areas to geocoded events from the Mapuche Conflict Event and Data Archive (MACEDA), focusing on attacks, coercive actions, and land invasions (Cayul et al., 2022). External exposure counts events occurring outside the receiving area at an exact minimum straight-line distance of at most 25 kilometers from its footprint; conflict inside the area enters separately. The 25-kilometer measure is intended to map nearby external exposure, not to identify the specific channel through which one location affects another. Geographic proximity can proxy for several forms of economic connectivity and does not by itself identify a transmission mechanism (Debarsy and Le Gallo, 2025). The baseline estimates compare changes within the same urban area over time, absorb region-specific annual shocks, condition on contemporaneous within-area conflict, and allow trajectories to differ with pre-sample population, rurality, and historical conflict exposure.

Greater external conflict exposure is associated with weaker formal business activity. An increase from the 25th to the 75th percentile of positive annual exposure is associated with 2.2 percent fewer registered enterprises and 5.3 percent lower tax-reported firm sales. The enterprise association also appears in the timing evidence: coefficients remain negative over the first several years after exposure, while future exposure does not systematically predict current enterprise outcomes. Sales dynamics are less precisely estimated. Fiscal appraisal, nighttime lights, and newly built area do not show a corresponding contemporaneous contraction. Fiscal appraisal turns negative at medium lags, consistent with its interpretation as a slowly updated assessed stock rather than a high-frequency market price. Administrative registration activity has a negative full-period association, while recorded terminations show no clear association in 2005–2015, before the administrative interventions of 2016–2017. These series do not identify how enterprise stocks adjust. Within-area grid comparisons show negative associations between exposure and selected legal-entity counts at 5 and 10 kilometers and commercial-use share at 10, 20, 25, and 30 kilometers; nighttime lights,

appraisal, and newly built area show no common negative gradient (Appendix D).

Historical territorial geography provides an additional distinction. The formal-business association does not become systematically stronger where *Títulos de Merced* cover more of the surrounding territory or where the 1992 Mapuche population share is higher. These results do not imply that historical claims or Indigenous presence are economically irrelevant. Rather, they provide no evidence that the negative association with contemporary conflict exposure is mechanically concentrated in places with greater historical Mapuche presence. The data also do not identify a single transmission channel. Security concerns, access to customers and suppliers, expectations about territorial disputes, and other forms of local connectivity remain plausible mechanisms.

The paper makes three contributions. First, it documents a spatial margin of territorial conflict that is difficult to observe from either conflict locations or regional aggregates alone: recurrent conflict outside an urban area is associated with formal economic activity inside nearby cities. Second, it adds a distinct conflict setting to a literature dominated by armed conflict, civil war, terrorism, and severe criminal violence. The Mapuche case is a persistent Indigenous self-determination conflict that operates alongside an active institutional architecture of property registration, Indigenous-land protection, and restitution. Third, by combining formal business records with directly spatial urban outcomes and historical land institutions, the paper shows that the formal-activity association is not accompanied by a common contemporaneous physical decline and is not systematically stronger where historical claims or baseline Mapuche presence are greater. More broadly, the results suggest that the economic incidence of territorial conflict may extend beyond the land under dispute, so assessments of its local consequences should also consider nearby urban economies.

2 Institutional Setting and Conflict Geography

2.1 Territorial self-determination and historical claims

The Mapuche are an Indigenous group of south-central Chile whose historical territory encompasses much of the study corridor. Mapuche communities retained substantial autonomy through most of the colonial period and were incorporated into the Chilean State only after the military occupation of Araucanía in the late nineteenth century. Between 1884 and 1929, the State confined Mapuche communities to nearly 3,000 reservations covering roughly 550,000 hectares, a small share of their ancestral territory (Jaimovich and Jordán, 2025b; Jaimovich and Toledo-Concha, 2026). The State formalized these reservations through *Títulos de Merced* (TDM). The TDM therefore did not restore ancestral territory; they legally recognized the much smaller areas left to Mapuche communities after the military occupation. They created an important State-recognized geography of historical territorial claims that remains relevant to later land protection, restitution, and conflict. Importantly, TDM boundaries do not describe current Mapuche ownership. Through subsequent allotment, land sales, encroachment, weak enforcement, and later deregulated transfers, Mapuche communities lost ownership of substantial amounts of land within the historical reservations (Jaimovich and Jordán, 2025b,a). Contemporary claims can therefore concern land that lies

within a historically recognized TDM but is no longer under Mapuche ownership. In the empirical analysis, TDM are used to locate this historical geography of recognized claims, not to identify the current owner of a parcel or the ethnic identity of a nearby firm.

Agrarian reform extended the contested geography beyond the reservations. The reform began in 1962 and accelerated between 1970 and 1973, when Mapuche communities recovered or gained access to land through the expropriation of large estates. After the 1973 military coup, the dictatorship returned or reassigned much of that land, promoted forestry plantations, and divided remaining collective reservations into individual parcels. Contemporary invasions and attacks are more common on plots touched by that reform and its reversal, linking current conflict to grievances both within and beyond TDM boundaries (Jaimovich and Jordán, 2025a; Jaimovich and Toledo-Concha, 2026).

2.2 Formal land institutions and restitution

The return to democracy in 1990 introduced a new legal framework for Indigenous land claims. The 1993 Indigenous Law recognizes Indigenous land through legal history, including TDM, long-standing occupation, judicial recognition, and land later transferred by the State. It restricts alienation and encumbrance, although the rules differ between community and individually held land and permit specified transactions with authorization from the *Corporación Nacional de Desarrollo Indígena* (CONADI), the public agency created by the same law to implement Indigenous policy, including the protection and expansion of Indigenous land (Gobierno de Chile, 1993). The law also created the Public Registry of Indigenous Territories, which makes registration central to the enforcement of Indigenous-land status. Evidence from the Temuco property records shows that registered parcels experienced fewer post-inheritance sales and smaller losses of Mapuche ownership, while registry coverage declined over time (Jordán et al., 2025). The law also created an institutional mechanism for addressing historical land claims. CONADI administers the Fund for Indigenous Land and Water, including the mechanism established under Article 20(b), through which the State acquires land to address recognized Indigenous land claims. Between 1993 and 2024, CONADI acquired 230,563 hectares through this mechanism in Biobío, Araucanía, Los Ríos, and Los Lagos. The 2025 Presidential Commission for Peace and Understanding estimated that existing claims and applications corresponded to approximately 396,825 hectares under its proposed parameters, implying that the State had acquired an amount equivalent to about 58 percent of this institutionalized demand. Including estimated future demand increases this figure to approximately 486,000 hectares. These estimates refer to claims processed or potentially processed through the formal restitution system rather than to the full extent of historical Mapuche territory. Land restitution is therefore not only a historical response to dispossession but an ongoing State institution during the period of contemporary conflict. Appendix G documents how recorded CONADI land acquisitions evolve with conflict exposure during the study period.

2.3 Conflict geography and recurrence

The Mapuche Conflict Event and Data Archive (MACEDA) records geocoded events from local-media reports in the Mapuche self-determination conflict (Cayul et al., 2022). The main event family contains attacks, coercive actions, and land invasions. Events recur over time rather than defining a single absorbing transition into conflict.

Most main events occur outside the study's urban areas. Of the 2,775 main events in the four-region corridor, 2,288 (82.5%) lie outside all 104 fixed urban areas. La Araucanía contains 66.6 percent of these events despite accounting for 25.7 percent of the corridor's 2017 population; conflict also occurs in Biobío, Los Ríos, and Los Lagos. Figure 1 maps the events and the locations of the receiving urban areas.

Annual event counts rise and fall throughout the period (Figure 2). Nearby urban areas may therefore face repeated changes in external exposure even when events occur beyond their own urban footprint.

3 Data

3.1 Spatial units and outcomes

The analysis follows 104 fixed urban areas in Biobío, Araucanía, Los Ríos, and Los Lagos. Each area is constructed from the 2017 INE urban census blocks within a municipality and remains fixed over time; its footprint can contain spatially disconnected components. These are geographic urban areas, not functional labor-market areas. Their median area is 4.59 km^2 (mean 8.21); the interquartile range is 2.45–7.60 km^2, and the full range is 0.61–63.91 km^2.

Annual records from Chile's national tax authority (Servicio de Impuestos Internos, SII) measure registered enterprises and tax-reported sales. Tax-reported sales are sales declared by firms that SII administratively assigns to the municipality of their tax or headquarters address. Both series are available from 2005 to 2021. These outcomes measure formal activity associated with each urban area; they do not establish that production or sales occur inside the fixed urban footprint.[1]

Nighttime luminosity provides a spatial measure of activity within the fixed urban footprints over a longer period than the SII business records. Lights do not measure formal firms or sales directly, but provide a complementary indicator of broader urban economic activity.

The property cadastre measures total fiscal appraisal and mean appraisal per property within each fixed urban geography. It sums assessed land and, where present, construction values to determine the property-tax base. It is an administratively updated assessed property stock, not a transaction or commercial price. SII updates appraisals through mass revaluations, individual inspections, taxpayer requests, and semiannual CPI indexation. The 2010–2021 panel includes nationally scheduled non-agricultural mass revaluations in 2013, 2014, and 2018; the 2006 round preceded the panel, and the 2013 and 2014 rounds covered different property classes. Annual

[1] Appendix E examines this spatial interpretation.

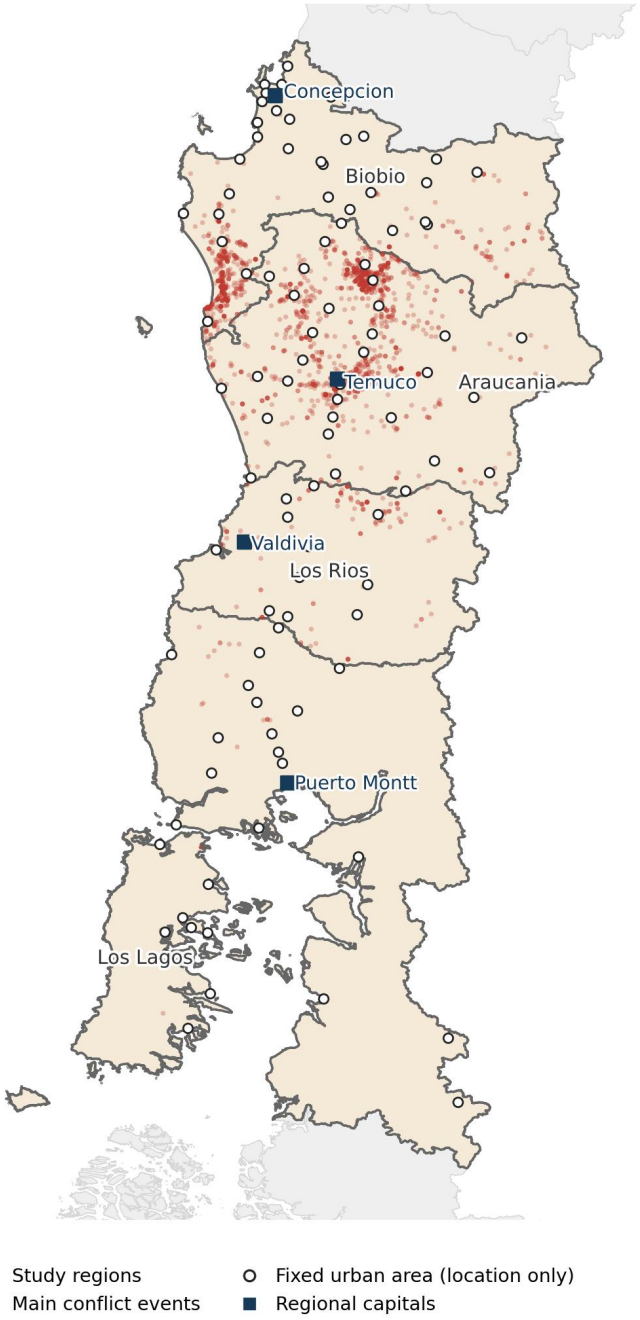


Figure 1: Main conflict events and fixed urban areas in southern Chile. The map shows attacks, coercive actions, and land invasions in Biobío, Araucanía, Los Ríos, and Los Lagos during 1990–2021. Point markers locate the 104 receiving fixed urban areas, with squares for the four regional capitals; marker size does not represent footprint area.

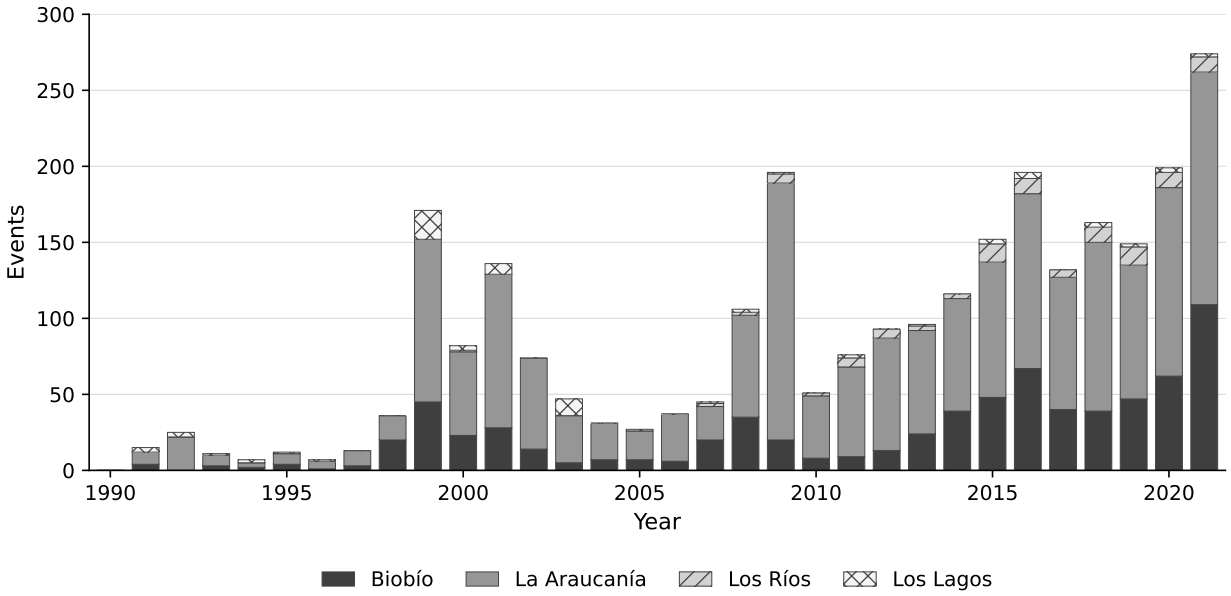


Figure 2: Annual main Mapuche conflict events by region, 1990–2021. Stacked bars count attacks, coercive actions, and land invasions in Biobío, Araucanía, Los Ríos, and Los Lagos.

changes and lagged estimates therefore need not track market-price responses; revaluation schedules, inspections, and indexation govern when assessed values change. Region-by-year effects absorb revaluation-year shocks shared within regions, but not heterogeneous parcel or cadastral changes.[2]

WSF Evolution records changes in built surface. It measures a physical flow and complements the economic outcomes by tracking built-area expansion. Its temporal support and interpretation differ from the headline outcomes, so detailed results appear in the appendix. Appendix A documents the construction, transformations, temporal coverage, and spatial support of each series.

3.2 Conflict data and external exposure

Conflict exposure comes from annual geocoded events in the Mapuche Conflict Event and Data Archive (MACEDA) (Cayul et al., 2022). The main event family includes attacks, coercive actions, and land invasions; protests are excluded. Let A_{mt} denote eligible events occurring within 25 kilometers of urban area m in year t, but outside the receiving area. External exposure is

$$D_{mt} = \log(1 + A_{mt}) \,. \tag{1}$$

Conflict counts are zero-heavy and right-skewed. The $\log(1 + x)$ transformation retains zero-exposure area-years and dampens the influence of high counts; coefficients therefore describe changes in transformed exposure and are not elasticities with respect to the raw event count. Appendix B applies the inverse hyperbolic sine consistently to contemporary external conflict, within-area conflict, and cumulative pre-2005 conflict, holding the sample and other controls fixed.

[2]See SII descriptions of fiscal appraisal and prior revaluation schedules (Servicio de Impuestos Internos, 2026a,c,b).

Distance is the exact minimum straight-line distance from each event point to the fixed urban-area footprint, calculated in metric projection EPSG:32718. Events outside the receiving area qualify when this distance is at most 25 km. The exposure includes events outside all study urban areas and events inside another study urban area; it is not restricted to events coded as rural. Events inside the receiving area enter separately as $I_{mt} = \log(1 + B_{mt})$, where B_{mt} is the annual count of main events inside that area.

The conflict literature does not provide a standardized distance cutoff for nearby exposure. Closely related studies use local windows in the 20–30 km range (Brancati et al., 2026; Fadare et al., 2024), while other work defines exposure along transport corridors (Alfano and Cornelissen, 2026; Liu et al., 2026). I use 25 km based on spatial and connectivity considerations, not outcome estimates, and assess alternative radii from 10 to 60 km. An outcome-free comparison finds that annual area-level exposure counts under the 20–30 km definitions are strongly correlated with counts under a 30-minute contemporary road-network definition (Appendix A.4). The road network uses imputed speeds for most road length and does not reconstruct historical travel over 1990–2021.

Because exposure is defined separately for each receiving area, an event may contribute to multiple areas when their windows overlap. Of 2,775 eligible events in 1990–2021, the preferred mapping contains 8,480 event–area pairs. Each event enters about 3.1 areas on average (median 3; maximum 9), while 109 events (3.9 percent) enter no area's external exposure. Events outside all study urban areas make up 81.7 percent of receiver–event pairs; events inside another study urban area make up the remainder. Each area-year remains one panel observation. Figure A.3 shows the event-level distribution. Both location categories contribute to the preferred exposure measure.

4 Empirical Strategy

4.1 Baseline specification and identifying variation

For outcome Y_{mt}, the preferred area-level specification is

$$Y_{mt} = \beta D_{mt} + \rho I_{mt} + \mu_m + \lambda_{r(m),t} + \tau_t (X_m^0)'\theta + \varepsilon_{mt}, \tag{2}$$

where X_m^0 contains standardized 2002 municipal log population, 2002 municipal rural share, and cumulative exact-25-km external conflict from 1990 through 2004. The terms μ_m and $\lambda_{r(m),t}$ are area and region-by-year fixed effects, and τ_t is linear calendar time. Standard errors are clustered by area. Because nearby areas can share exposure and local shocks, Appendix B also reports spatial-HAC inference allowing spatial and temporal residual dependence.

The preferred specification allows secular trajectories to differ with initial population, initial rural composition, and pre-sample external conflict exposure. Population and rural share are measured for the municipality in the 2002 Census, not within the later fixed urban footprint. Population enters as $\log(1 + P_m^{2002})$. Historical conflict uses the same broad-external exact-25-km definition as contemporary exposure, summing eligible events from 1990 through 2004 and then

taking $\log(1 + H_m)$. These interactions allow areas with different initial scale, rural composition, and prior conflict geography to follow different linear trajectories. Appendix B documents their construction.

Region-by-year effects absorb annual shocks shared by areas within Biobío, Araucanía, Los Ríos, and Los Lagos, including regional economic and policy conditions. The model also conditions on contemporaneous conflict inside the receiving area. The external coefficient therefore captures the association with nearby conflict conditional on direct local conflict; this co-exposure is not assumed to be exogenous.

The characteristics predate the 2005–2021 formal-activity and 2010–2021 appraisal panels. They do not predate the full lights and built-area windows, beginning in 1992 and 1990. Those longer-window models retain area and region-by-year effects and within-area conflict but omit all three trends.

Unobserved local shocks could affect both conflict and economic activity. Area effects remove persistent differences, region-by-year effects remove shared regional shocks, and the trends allow trajectories to differ with initial conditions. The coefficient β uses the remaining within-area variation in exposure. It is a conditional association because this adjustment does not make subsequent conflict random.

4.2 Timing patterns

The timing analysis extends the preferred specification by including future, contemporaneous, and lagged values of the same annual exposure measure:

$$\begin{aligned} Y_{mt} = &\sum_{h=1}^{3} \pi_h D_{m,t+h} + \sum_{\ell=0}^{L_Y} \beta_\ell D_{m,t-\ell} + \rho I_{mt} + \mu_m + \lambda_{r(m),t} \\ &+ \tau_t (X_m^0)' \theta + \varepsilon_{mt}. \end{aligned} \tag{3}$$

The preferred projection includes three future-exposure terms and lags through eight years for enterprises, sales, and fiscal appraisal. The nighttime-lights profile extends through ten years because of its longer temporal support.

For presentation, coefficients on future exposure are plotted at negative horizons. These terms show whether outcomes tend to move before the corresponding exposure is realized, while the contemporaneous and lagged coefficients trace how the association evolves during and after exposure. Because conflict exposure is continuous and recurrent, the exercise is best read as a description of its temporal pattern rather than as a conventional event study.

Appendix C reports additional timing evidence based on a past-window definition of major conflict escalations and estimates a non-absorbing difference-in-differences design (de Chaisemartin and D'Haultfœuille, 2024).

5 Results

5.1 Baseline estimates

Nearby external conflict is associated with lower formal business activity. In the preferred specification, moving from the 25th to the 75th percentile of positive exposure corresponds to about 2.2 percent fewer registered enterprises and 5.3 percent lower tax-reported sales (Table 1).

Table 1: External conflict exposure and urban-area outcomes

Specification	(1) Registered enterprises	(2) Tax-reported sales	(3) Nighttime lights	(4) Fiscal appraisal total	(5) WSF new built area
Area + year FE	-0.020**	-0.050**	-0.005	0.001	-0.041
	(0.009)	(0.022)	(0.011)	(0.021)	(0.034)
+ predetermined trends	-0.012**	-0.034**	—	0.002	—
	(0.006)	(0.017)		(0.016)	
+ region-by-year FE	-0.013**	-0.032*	-0.004	0.019	0.018
	(0.007)	(0.019)	(0.012)	(0.013)	(0.031)
+ within-area conflict (preferred)	-0.013**	-0.032*	-0.004	0.019	0.021
	(0.006)	(0.019)	(0.012)	(0.013)	(0.030)
Observations	1,768	1,768	3,060	1,248	2,704
Areas	104	104	104	104	104
Sample period	2005–2021	2005–2021	1992–2021	2010–2021	1990–2015

Notes: External exposure counts attacks, coercive actions, and land invasions within 25 km of the fixed urban footprint, excluding events inside the receiving area. All specifications include area fixed effects. The four rows add year effects, the three predetermined trends, region-by-year effects, and contemporaneous within-area conflict. The trends interact calendar time with 2002 municipal population, 2002 rural share, and cumulative external conflict in 1990–2004. Predetermined trends are not estimated for nighttime lights and WSF because their panels begin before the baseline characteristics. WSF is measured in asinh-transformed hectares. Standard errors are clustered by fixed urban area. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The enterprise and sales associations remain negative as the specification adds predetermined trends, region-by-year effects, and within-area conflict. The trends allow outcome paths to differ with initial population, rural composition, and prior conflict. The contemporaneous estimates for lights, appraisal total, and new built area show no common contraction: lights are close to zero, while appraisal and built area have positive, imprecise estimates.

The enterprise and sales estimates remain negative across exposure radii from 10 to 60 km, including 20, 25, and 30 km (Appendix B). These specifications vary the current exposure radius while holding the historical trend at 25 km.

The negative enterprise association is not confined to firms classified in forestry or wood and paper manufacturing. Excluding those activities changes the coefficient from -0.0132 to -0.0127; sales estimates also remain negative under both constructions in Table B.6. The exclusion estimates are less precise, and retained firms may still depend on forestry demand (Appendix B.6).

5.2 Timing patterns

The continuous timing profile shows a persistent negative enterprise association over the first several years after exposure, with the strongest pattern around the medium-term lags (Figure 3). Future exposure does not show a comparable systematic enterprise pattern. That absence does not show that conflict is unanticipated or exogenous. Because exposure recurs and remains continuous, these lead-lag estimates describe timing rather than isolate a one-time treatment effect.

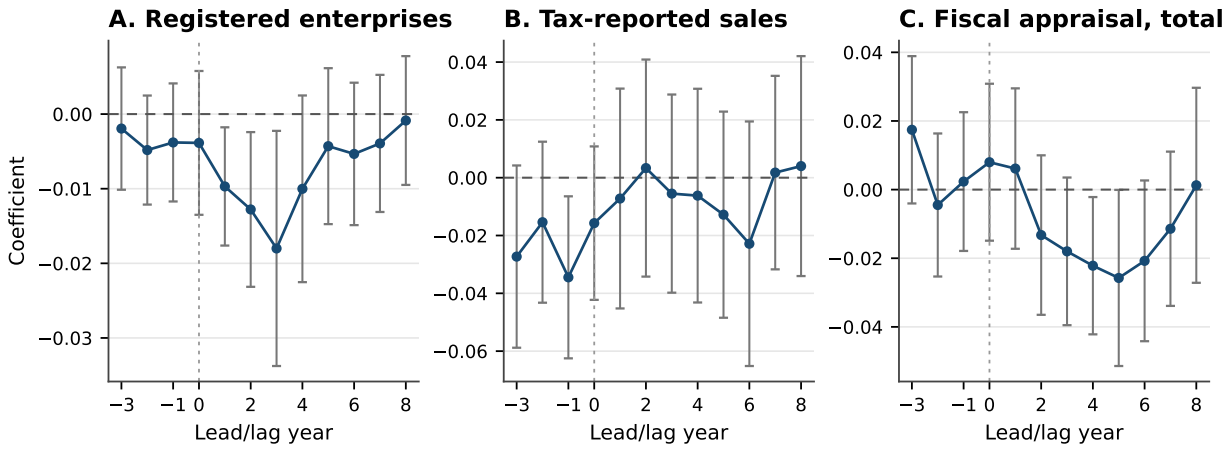


Figure 3: Timing patterns in formal activity and fiscal appraisal. The panels report coefficients from the continuous lead-lag projection defined in Section 4. Negative values on the horizontal axis are future exposure terms, zero is current exposure, and positive values are lagged exposure terms. Vertical bars are 95 percent confidence intervals based on standard errors clustered by fixed urban area. All specifications use broad external exposure within 25 kilometers, condition on contemporaneous within-area conflict, and include area and region-by-year effects and the three predetermined trends defined in Section 4.1.

Sales estimates are generally negative, but their uncertainty leaves no clear dynamic profile. Fiscal appraisal does not fall contemporaneously; its negative pattern emerges at medium lags. Because appraisal records an administratively updated property stock, that delay may reflect when values are reassessed, and does not by itself establish a delayed causal effect.

Because nearby conflict exposure recurs rather than beginning at a single date, the continuous profile remains the main timing evidence. The spike exercise asks whether unusually large discrete escalations produce a clearer response, using a non-absorbing DiD design. Neither registered enterprises nor tax-reported sales show a clear contraction after a spike. Fiscal appraisal estimates turn negative at longer horizons, but limited support and placebo movements make that pattern difficult to interpret. The spike estimates therefore remain secondary evidence (Appendix C.2).

5.3 Spatial interpretation and supporting evidence

SII outcomes are assigned through municipal tax and headquarters addresses, so they do not by themselves locate all reported activity inside urban areas. For the selected legal entities that

can be matched to cadastral headquarters, the negative association is also visible among urban-headquartered entities (Appendix E.1).

The grid comparison asks whether urban cells with greater external exposure also show weaker local economic activity or land use than other cells in the same area and year. More exposed cells have fewer selected legal entities and lower commercial-use shares at several radii, while lights, appraisal, and new built area show no common gradient (Appendix D). These within-area differences do not measure aggregate urban decline or establish that activity moved between cells.

If the municipal SII association mainly reflects firms tied to the surrounding rural economy, sectors with stronger predetermined agricultural or forestry input and customer linkages should respond more strongly. The estimates show no systematic gradient by these linkages (Appendix E).

6 Additional Evidence

6.1 Robustness

The enterprise and sales estimates remain negative as the model moves from common year effects to region-by-year and province-by-year effects, although the finer geographic controls absorb more exposure variation and reduce precision (Appendix B). Applying the asinh transformation consistently to conflict measures produces similar enterprise and sales magnitudes for the same change in raw exposure.

Excluding aquaculture, the sector most directly exposed to the 2007–2010 Los Lagos crisis (Ritter, 2025), leaves the enterprise estimate almost unchanged. In the common area-years with subsector-sales data, the sales estimate remains negative but imprecise. Alternative wild-cluster and spatial-HAC inference supports the same negative formal-activity pattern, though uncertainty varies with the assumed dependence across areas. These procedures do not address confounding.

The formal-activity association remains similar after controlling for nearby protests and protests inside the receiving area. This makes it less likely that the baseline estimates simply reflect broader episodes of Mapuche-related mobilization (Appendix B.2, Table B.2).

The estimates do not show a systematically stronger association in areas with greater historical *Títulos de Merced* coverage or a larger 1992 Mapuche population share (Appendix F).

6.2 Margins of adjustment

The enterprise stock and administrative registration activity are more negatively associated with nearby conflict at transport locations than with conflict at other locations. The registration difference is similar in magnitude when exposure is restricted to attacks, although less precisely estimated. Recorded terminations show no clear association with nearby conflict in 2005–2015, before SII administrative actions increased terminations in 2016–2017 (Appendices B.8 and B.9). The negative full-period registration estimate is also sensitive to the observation window. These administrative series do not identify the contribution of entry and exit to the lower enterprise stock.

7 Discussion and Conclusion

This paper studies whether persistent territorial conflict is associated with weaker business activity in nearby urban areas, even when events occur outside the receiving area. In southern Chile, an increase from the 25th to the 75th percentile of positive external exposure corresponds to 2.2 percent fewer registered enterprises and 5.3 percent lower tax-reported sales. These associations account for regional annual shocks, three pre-sample characteristics, and contemporaneous conflict inside the receiving area.

The results do not show a common contemporaneous decline in urban lights, fiscal appraisal, and newly built area. Administrative registration activity has a negative full-period association; recorded terminations show no clear association before the 2016–2017 administrative break. Negative appraisal coefficients appear at medium lags in the continuous projection, although placebo movements limit the interpretation of the escalation estimates. The formal-activity association is also not systematically stronger near historically recognized claims or in areas with a higher 1992 Mapuche population share.

Firms could respond to nearby conflict through changes in security, access to customers and suppliers, or expectations about land use and territorial disputes. The data do not distinguish these channels. The enterprise-stock and first-registration associations are more negative for transport-location conflict, while sales show no comparable concentration and the transport–nontransport difference for recorded terminations is imprecisely estimated. This pattern is consistent with access disruption or transport-related insecurity weakening formal registration activity. These comparisons do not identify transport disruption as a causal mechanism. State coercive responses do not appear to be associated with an offsetting improvement in nearby formal activity (Appendix B.2). State responses may occur where conflict is more severe, so these conditional associations do not identify the causal effect of State intervention.

The negative association is also visible among selected legal entities headquartered inside urban areas. Within the same urban area and year, relatively more exposed grid cells have weaker selected legal-entity presence and lower commercial-use shares. The sectoral input-output analysis finds no stronger association in sectors with greater predetermined rural production linkages. These results make it less likely that the municipality-level SII association simply reflects activity in the surrounding rural economy.

Regional annual effects and pre-sample conflict trends leave feedback and unobserved confounding unresolved. Municipal tax addresses also limit how precisely SII activity can be located, and the longer physical series require different trend controls because their windows begin before the baseline characteristics. Within these limits, the paper documents an economic reach beyond conflict locations that is more visible in formal business activity than in contemporaneous physical change. That reach can coexist with functioning property, tax, land-registration, and restitution institutions.

The observed CONADI response raises the question whether land subsidies under Article 20(a) or claim-related purchases under Article 20(b) subsequently reduce conflict and its associated eco-

nomic costs. Appendix G combines these programs and does not identify either effect.

References

Albertus, M. (2026). Adhering Indigenous Communities to the State: Recognition Politics During Civil Conflict.

Alfano, M. and Cornelissen, T. (2026). Spatial Spillovers of Conflict in Somalia. *American Economic Review*, 116(6):2166–2201.

Besley, T. and Mueller, H. (2012). Estimating the Peace Dividend: The Impact of Violence on House Prices in Northern Ireland. *American Economic Review*, 102(2):810–833.

Brancati, E., Di Maio, M., Gatti, R., and Islam, A. M. (2026). Under pressure: Global evidence on conflict and firms. *Journal of Development Economics*, 182:103807.

Camacho, A. and Rodriguez, C. (2013). Firm Exit and Armed Conflict in Colombia. *Journal of Conflict Resolution*, 57(1):89–116.

Cayul, P., Corvalan, A., Jaimovich, D., and Pazzona, M. (2022). Introducing MACEDA: New micro-data on an indigenous self-determination conflict. *Journal of Peace Research*, 59(6):903–912.

Comisión Verdad Histórica y Nuevo Trato con los Pueblos Indígenas (2008). *Informe de la Comisión Verdad Histórica y Nuevo Trato con los Pueblos Indígenas*. Comisionado Presidencial para Asuntos Indígenas, Santiago de Chile.

de Chaisemartin, C. and D'Haultfœuille, X. (2024). Difference-in-differences estimators of intertemporal treatment effects. *The Review of Economics and Statistics*, pages 1–45. Advance online publication.

Debarsy, N. and Le Gallo, J. (2025). Identification of spatial spillovers: Do's and don'ts. *Journal of Economic Surveys*, 39(5):2152–2173.

Del Prete, D., Di Maio, M., and Rahman, A. (2023). Firms amid conflict: Performance, production inputs, and market competition. *Journal of Development Economics*, 164:103143.

Fadare, O., Srinivasan, C. S., and Zanello, G. (2024). Livestock diversification mitigates the impact of farmer–herder conflicts on animal-source foods consumption in nigeria. *Food Policy*, 122:102586.

Feir, D. L., Gillezeau, R., and Jones, M. E. (2023). Institutional Drift, Property Rights, and Economic Development: Evidence from Historical Treaties.

Gobierno de Chile (1993). Ley N° 19.253: Establece normas sobre protección, fomento y desarrollo de los indígenas, y crea la Corporación Nacional de Desarrollo Indígena. Diario Oficial.

Jaimovich, D. and Jordán, F. (2025a). Indigenous Property and Economic Development Under the Pinochet Dictatorship. In González, F. and Prem, M., editors, *The Pinochet Shock: Radical Economic Change and Life Under Dictatorship in Chile*, pages 317–348. Springer Nature Switzerland, Cham.

Jaimovich, D. and Jordán, F. (2025b). The Long Shadow of Property Rights in the Mapuche Homeland, Chile.

Jaimovich, D., Jordán, F., Black, M., and Heilmayr, R. (2025). Environmental Impacts of Indigenous Land Restitution in Chile.

Jaimovich, D. and Toledo-Concha, F. (2026). The grievances of a failed reform: Chilean land reform and conflict with indigenous communities. *The Economic Journal*, page ueag029.

Jordán, F., Jaimovich, D., and Heilmayr, R. (2025). The Impacts of Restrictions to Individual Rights on Indigenous Lands. *IDB Publications*.

Larroulet Vignau, C. and Sepúlveda, J. P. (2025). The economic impact of violence in southern Chile. *Applied Economics*, 57(54):9203–9214.

Liu, J., Olabisi, M., and Maredia, M. K. (2026). Market integration and the impact of violence on trade networks: Evidence from nigeria. *European Review of Agricultural Economics*, 53(3):1000–1053.

Neri-Lainé, M. (2026). Firms under fire! How insecurity affects formal firms' existence. *Journal of Development Economics*, 180:103714.

Ritter, S. (2025). The economic effect of splitting a region in a centralized country: A case from chile. *Global Challenges & Regional Science*, 2:100011.

Servicio de Impuestos Internos (2026a). Avalúo fiscal de bienes raíces. `https://www.sii.cl/destacados/impuesto_territorial/avaluo_fiscal.html`. Accessed October 2, 2026.

Servicio de Impuestos Internos (2026b). Reajustes y exenciones de impuesto territorial. `https://www.sii.cl/ayudas/ayudas_por_servicios/2242-reajustes_exenciones-2468.html`. Accessed May 18, 2026.

Servicio de Impuestos Internos (2026c). Reavalúos anteriores. `https://www.sii.cl/destacados/reavaluo/reavaluos_anteriores.html`. Accessed October 2, 2026.

Utar, H. (2025). Firms and labor in times of violence: Evidence from the mexican drug war. *The World Bank Economic Review*, 39(3):632–662.

A Data, measurement, and sample construction

A.1 Fixed urban-area geography and study sample

The analysis follows 104 fixed urban areas in Biobío, Araucanía, Los Ríos, and Los Lagos. Each area is constructed from the 2017 INE urban census blocks ("manzanas" with `AREA=1`) within a municipality and is held fixed over time; its footprint can contain spatially disconnected components. The four regions cover the part of south-central Chile where historical Mapuche occupation, recognized claims, contemporary restitution, and recent conflict most clearly overlap. The sample is selected for this overlap and is not nationally representative.

SII enterprise and sales records are observed at municipal tax addresses and mapped to the corresponding urban area; they therefore measure formal activity associated with the area rather than activity mechanically located inside its urban footprint.

Figure A.1 provides historical and demographic context for the study corridor.

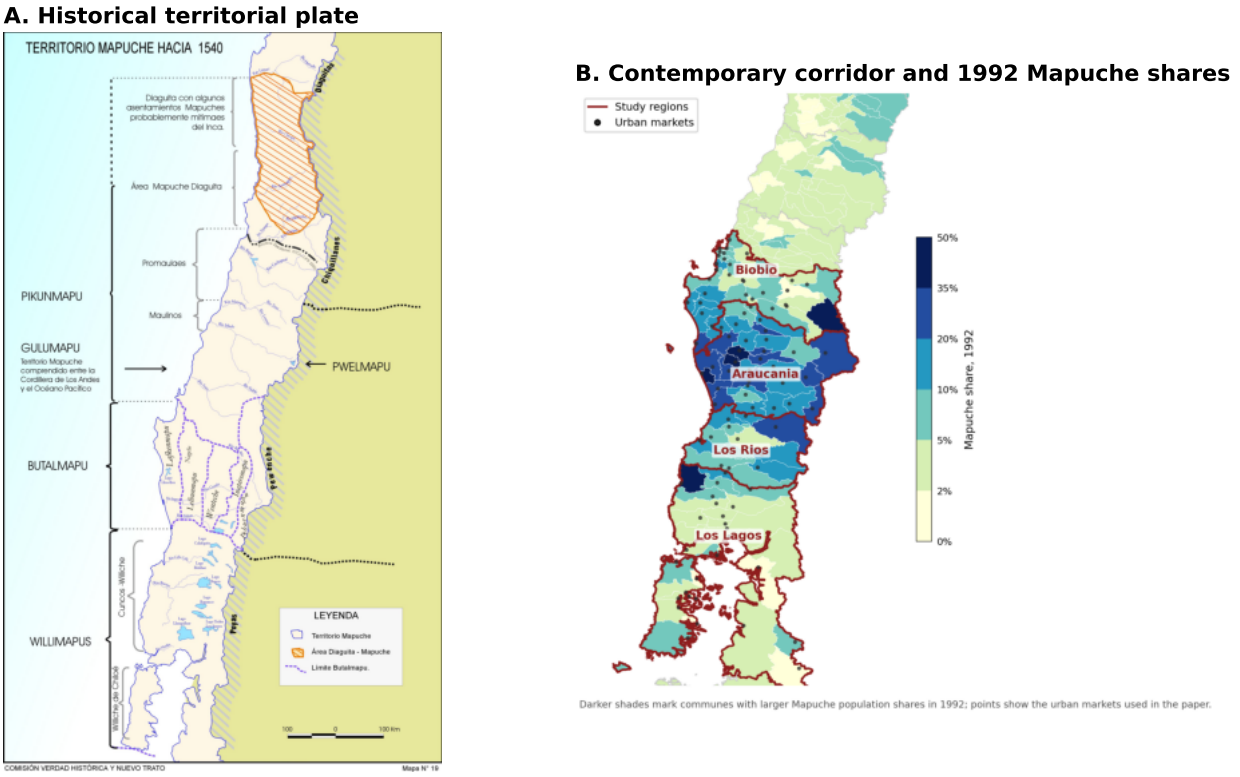


Figure A.1: Historical Mapuche territory and the contemporary study corridor. Panel A reproduces a historical map of Mapuche territorial organization around 1540 from the Comisión Verdad Histórica y Nuevo Trato. Panel B shows the four-region study corridor and Mapuche population shares in 1992. Together, the panels illustrate the historical and demographic geography underlying the study area.

The business panel covers 2005–2021. Fiscal appraisal begins in 2010, while harmonized nighttime lights and WSF built-area data have longer, but distinct, temporal support. Table A.2 records

the construction and transformation of each outcome used in the analysis.

A.2 Conflict data and event composition

Conflict events come from MACEDA. The main analysis counts attacks, State coercive actions, and land invasions; protests are excluded. This family contains 2,775 events, of which 82.1 percent have MACEDA's rural classification and 82.5 percent lie outside all fixed urban areas used to construct the exposure measure. Most events are therefore rural, but an event qualifies as external exposure when it occurs outside the receiving urban area.

Table A.1: MACEDA event composition and regional concentration

Event family	Events	Rural MACEDA	Share	
Panel A. Event composition				
All MACEDA events	3,730	2,587	69.4%	
Main event family: attacks, coercion, land invasions	2,775	2,278	82.1%	
Protests	955	309	32.4%	
Panel B. Operational fixed-system geography of the main event family				
Classification	Events	Share		
Outside all fixed urban systems	2,288	82.5%		
Inside a fixed urban system	487	17.5%		
Panel C. Regional concentration of main events				
Region	Main events	Event share	2017 population	Population share
Biobio	752	27.1%	1,556,805	41.8%
La Araucanía	1,847	66.6%	957,224	25.7%
Los Lagos	76	2.7%	828,708	22.2%
Los Ríos	100	3.6%	384,837	10.3%
Panel D. La Araucanía share of cumulative main events				
Window	Main events	La Araucanía events	Share	
1990–1992	40	30	75.0%	
1990–2002	589	401	68.1%	
1990–2017	1,990	1,371	68.9%	
1990–2021	2,775	1,847	66.6%	

Notes: The main event family comprises attacks, coercive actions, and land invasions. Rural is MACEDA's event classification. The outside-area classification uses the fixed urban geography. Protests are not used in the exposure measure.

A.3 External exposure construction

For area m and year t, external exposure is $D_{mt} = \log(1 + A_{mt})$, where A_{mt} counts eligible events within 25 kilometers of the receiving area and outside that area. It includes events outside all fixed urban areas and events inside another fixed urban area. Events inside the receiving area enter separately as annual within-area conflict, with no radius cutoff.

Figure A.2 illustrates the exposure window around selected urban areas, and Figure A.3 shows how many receiving areas each event enters.

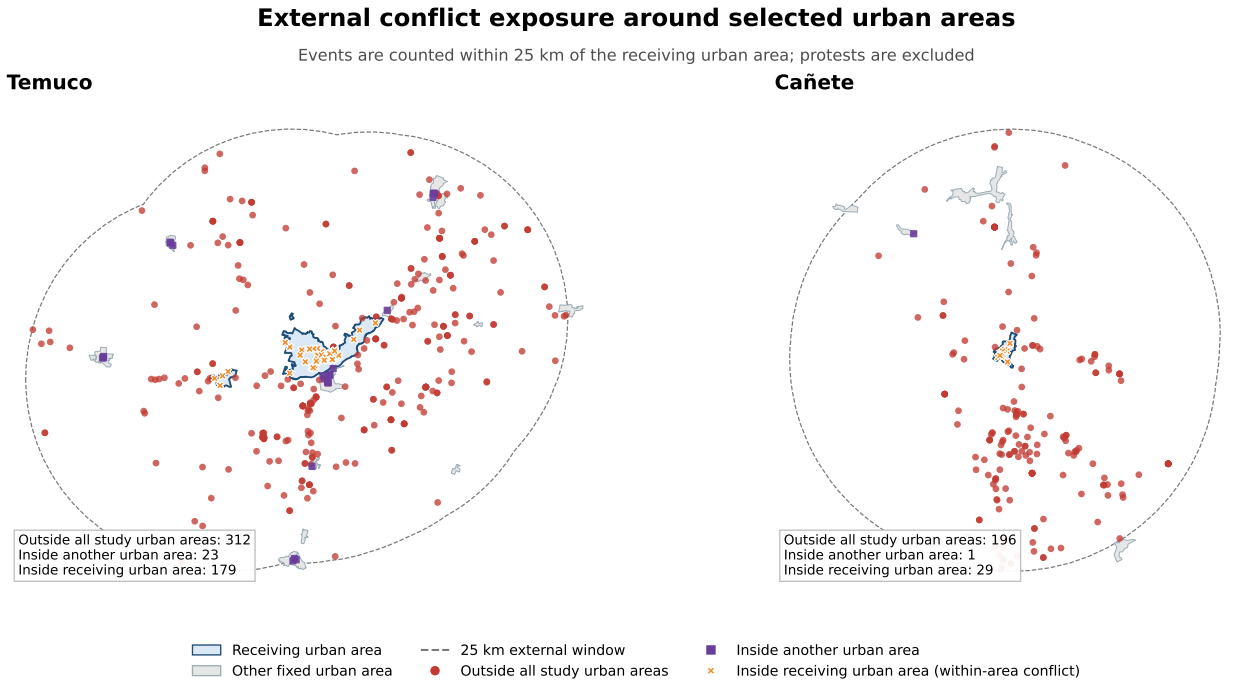


Figure A.2: External conflict exposure around selected urban areas. The dashed boundary marks the 25-kilometer window around each receiving area. Events outside all study urban areas and events inside another study urban area contribute to external exposure. Events inside the receiving area are shown separately as within-area conflict.

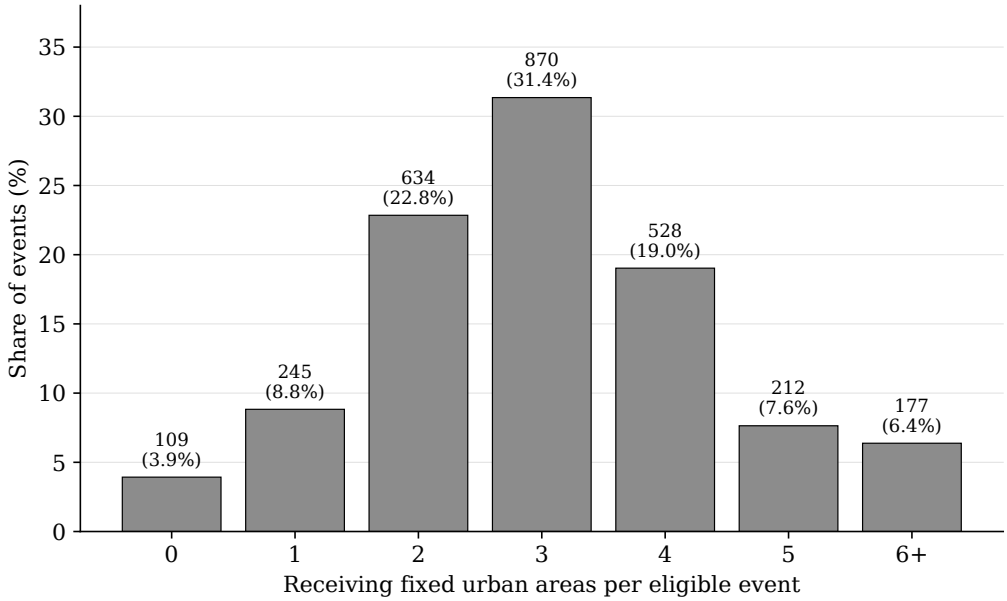


Figure A.3: Event multiplicity under exact 25 km external exposure. Bars show the share of 2,775 eligible MACEDA events entering the exposure of 0, 1, 2, 3, 4, 5, or at least 6 receiving fixed urban areas. Labels report counts and shares. Own-area events are excluded from that area's external count, but may enter another area's exposure.

A.4 Geographic distance and road accessibility

This outcome-free diagnostic compares exact event-point-to-footprint distance with fastest-route travel time on a contemporary road network. The September 2026 OpenStreetMap routing graph contains 122,772 km of motor-vehicle roads; numeric posted speeds cover 14.99 percent of road length in this routing crop, and class-based speeds are assigned to the remaining 85.01 percent. All 188 components of the 104 urban footprints serve as origins. Events are snapped to the nearest road segment, with a 2-km maximum; 22 events fail this cutoff or fall outside the routing crop and are omitted from road-time assignments. Ferries are excluded.

Definition	Event–area pairs	Jaccard vs. road	Annual-count corr.	Mismatch (% / %)*
Euclidean 20 km	6,614	0.576	0.905	38.0 / 10.8%
Euclidean 25 km	8,480	0.628	0.925	27.1 / 18.2%
Euclidean 30 km	11,455	0.640	0.922	13.9 / 28.5%
OSM road 30 min	9,513	—	—	—

*Both mismatch quantities are percentages: road pairs missed by the Euclidean definition / Euclidean pairs excluded by the road definition. Annual-count correlations use area-year counts for 1990–2021.

The annual counts correlate most with the 25-km definition, while pair-level Jaccard agreement is highest for 30 km. Neither comparison identifies an optimal radius, and Euclidean distance is not equivalent to driving time. These assignments describe spatial connectivity only: the network postdates the 1990–2021 event panel, speeds are partly imputed, and historical travel times are not reconstructed. The diagnostic contains no outcomes and tests no economic mechanism.

A.5 Outcome construction and SII measurement

Nighttime lights, fiscal appraisal, and WSF are extracted from fixed urban footprints. Fiscal appraisal is an assessed administrative stock, not a transaction price, and is updated more slowly than a high-frequency market price. WSF records newly built area and therefore measures a physical flow. For registered enterprises and tax-reported sales, the SII address is a tax or headquarters address; it does not identify where a firm produces or where its sales take place. Appendix E examines the implication of this distinction.

A.6 Exposure support

Table A.3 describes the support for external and within-area exposure in the full 1990–2021 area panel. Within-area conflict captures a distinct contemporaneous geography from the external-exposure measure.

Table A.2: Outcome construction and interpretation

Outcome	Source and spatial unit	Years	Transformation	Interpretation
Registered enterprises	SII municipality tax-address measure mapped to area	2005–2021	log(enterprises)	Formal enterprise stock associated with the area
Tax-reported sales	SII municipality tax-address measure mapped to area	2005–2021	log(sales)	Scale of formal activity associated with the area
Nighttime lights	Harmonized DMSP–VIIRS, fixed urban footprint	1992–2021	log(total DN)	Urban luminosity
Fiscal appraisal total	SII cadastre, fixed urban footprint	2010–2021	log(appraised value)	Assessed urban property stock
Fiscal appraisal mean	SII cadastre, fixed urban footprint	2010–2021	log(mean appraisal)	Mean assessed property value
WSF new built area	WSF Evolution, fixed urban footprint	1990–2015	asinh(new hectares)	New built-area flow

Notes: Registered enterprises and tax-reported sales are SII municipality tax-address measures mapped to urban areas. Nighttime lights, fiscal appraisal, and WSF are constructed within fixed urban footprints. Transformations follow each outcome's support and unit; they are not all logs.

Table A.3: Support for external and within-area conflict exposure

Exposure measure	Mean	SD	Positive area-years
External exposure, D_{mt}	0.556	0.947	33.6%
Within-area exposure, I_{mt}	0.063	0.283	5.9%
$\text{Corr}(D_{mt}, I_{mt}) = 0.374$; both positive: 4.7%			

Notes: $D_{mt} = \log(1 + A_{mt})$ is broad external exposure; I_{mt} is the log of one plus annual main conflict events inside the receiving area. The panel contains 104 areas from 1990 to 2021.

B Robustness and margins of adjustment

B.1 Where external conflict occurs

External exposure can include events outside every study urban area or events inside another study urban area but outside the receiving city. Events in another city still contribute to the receiving area's nearby exposure, so the comparison asks whether those events account for the main association. About 81.7 percent of external event-area pairs originate outside all study urban areas. Table B.1 reports these shares and enters the two locations jointly in the preferred specification. The enterprise association remains negative for events outside all study cities. For sales, both coefficients are negative, but the outside-city estimate is imprecise while the inside-another-city estimate is more precise. The equality tests do not distinguish the two coefficients, but they do not establish that the components have equal effects. The evidence therefore weighs against an other-city-only explanation for enterprises, while leaving the location contribution to sales unresolved.

Table B.1: Where external conflict occurs

Panel A. Exposure composition, 1990–2021	Outside all study urban areas	Inside another study urban area	All external events
Event–area pairs	6,931	1,549	8,480
Share of pairs	81.7%	18.3%	100.0%
Area-years with positive exposure	28.2%	14.6%	33.6%
Panel B. Joint exposure estimates, 2005–2021	Outside all study urban areas	Inside another study urban area	Equality p-value
Registered enterprises	-0.0139*	-0.0029	
	(0.0074)	(0.0062)	0.242
Tax-reported sales	-0.0162	-0.0498**	
	(0.0218)	(0.0225)	0.275

Notes: Eligible events are attacks, coercive actions, and land invasions outside the receiving area and within 25 km. Panel A counts event–area pairs in 1990–2021 and reports the share of area-years with positive exposure. The two component exposures have a correlation of 0.449. Panel B enters both locations jointly in 2005–2021 with area and region-by-year effects, the three predetermined trends, and contemporaneous within-area conflict. Standard errors are clustered by fixed urban area. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.2 Conflict composition and protest exposure

Exposure to attacks, coercive actions, and land invasions is strongly correlated with nearby protest exposure (0.718 across area-years in 2005–2021). Yet controlling jointly for external protests and protests inside the receiving area leaves the enterprise and sales coefficients on main-family exposure close to their baseline estimates (Table B.2). Protest coefficients are imprecise; adding a pre-2005 protest-history trend makes the enterprise protest estimate more negative without materially changing the main-family estimates. The coefficient comparisons do not establish different associations for conflict and protests.

MACEDA codes attacks and land invasions overwhelmingly to Mapuche actors and coercive actions overwhelmingly to the State. Among State-coded coercive actions in 2005–2021, Carabineros account for 361 of 487 events, followed by the judiciary (91) and PDI (27). Coercive actions include raids, repression, arrests, evictions, convictions, indictments, and coercive threats. When the source does not identify the initiator, MACEDA defaults to ACTOR1; these coded shares describe event-family composition, not verified initiation in every case.

In the joint two-group model, coercive-action estimates are negative for both formal outcomes, especially sales; the enterprise interval still permits a modest positive association. The contentious-action and coercive-action coefficients are not statistically distinguished. State responses may occur where underlying conflict is more severe, so these estimates do not identify the effect of State intervention.

Table B.2: Conflict composition, protest exposure, and formal activity

Exposure / specification	Registered enterprises	Tax-reported sales
Panel A. Main conflict and protest exposure		
Baseline: main-family exposure	-0.0132**	-0.0320*
	(0.0065)	(0.0189)
Protest-only exposure	-0.0073	-0.0012
	(0.0061)	(0.0199)
Joint: main-family exposure	-0.0126*	-0.0362*
	(0.0065)	(0.0195)
Joint: protest exposure	-0.0026	0.0124
	(0.0059)	(0.0202)
Joint + protest-history trend: main family	-0.0143**	-0.0374*
	(0.0065)	(0.0193)
Joint + protest-history trend: protests	-0.0094	0.0073
	(0.0061)	(0.0201)
Panel B. Contentious and coercive actions		
Contentious actions	-0.0140*	-0.0103
	(0.0077)	(0.0246)
Coercive actions	-0.0033	-0.0424**
	(0.0065)	(0.0208)
Equality of exposure coefficients (p)	0.296	0.393
Area-years	1,768	1,768
Urban areas	104	104

Notes: The sample covers 104 fixed urban areas in 2005–2021 (1,768 area-years). Outcomes and exposures are in logs; event counts use $\log(1+x)$. External events occur outside the receiving area and within exact 25-km point-to-footprint distance. All models include area and region-by-year effects and the three preferred predetermined trends, with standard errors clustered by area. The baseline controls for contemporaneous within-area main-family conflict. Protest-only and joint protest models control for both contemporaneous within-area main-family conflict and within-area protests. The additional protest-history trend uses pre-2005 external protests. Panel B enters external contentious and coercive actions jointly with their matching within-area exposures; contentious actions sum attack and land-invasion raw counts before $\log(1+x)$. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3 Transformation and inference sensitivity

The preferred model includes linear trends based on 2002 municipal population and rural share and cumulative external conflict in 1990–2004 at exact 25 km.

Table B.3 replaces $\log(1+x)$ with asinh for contemporary external and within-area counts and the cumulative pre-2005 count before its trend interaction. Sample and other controls are fixed. A common increase from one to ten events gives similar enterprise and sales associations; raw coefficients are not directly comparable.

Table B.3: Comparable raw-count contrasts under consistent conflict transformations

Outcome	Transformation	1 to 10 events (%)	95% CI	p
Registered enterprises	$\text{Log}(1+x)$	-2.22	[-4.34, -0.06]	0.045
Registered enterprises	Consistent asinh	-2.13	[-4.12, -0.10]	0.040
Tax-reported sales	$\text{Log}(1+x)$	-5.30	[-11.17, 0.95]	0.094
Tax-reported sales	Consistent asinh	-5.53	[-11.19, 0.50]	0.071

Notes: Percent associations equal $100\{\exp[\hat{\beta}(g(10)-g(1))]-1\}$, with g equal to $\log(1+x)$ or asinh. Outcomes are in logs. All three conflict quantities use the stated transformation. Both models use 1,768 area-years and area-clustered intervals.

Wild-cluster inference gives $p = 0.041$ for enterprises and $p = 0.079$ for sales (Table B.4). At the 60 km spatial cutoff and three annual lags, spatial-HAC gives $p = 0.013$ and $p = 0.019$. The 30 and 90 km cutoffs give similar dependence-adjusted precision. HAC uses a normal reference distribution; area-clustered headline intervals use t with 103 degrees of freedom. Neither addresses endogeneity.

Table B.4: Alternative inference for headline formal outcomes

Outcome	Coef.	Clustered SE	Wild p	Spatial-HAC SE	Spatial-HAC p
Registered enterprises	-0.0132	(0.0065)	0.041	(0.0053)	0.013
Tax-reported sales	-0.0320	(0.0189)	0.079	(0.0136)	0.019

Notes: Wild-cluster p-values use 999 repetitions and area clustering. Spatial-HAC standard errors use a 60-kilometer spatial cutoff and a three-year lag cutoff; 30- and 90-kilometer cutoffs give the same qualitative conclusion. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.4 Province influence

Omitting each province in turn leaves the enterprise and sales estimates negative, although precision varies with the retained sample (Figure B.1).

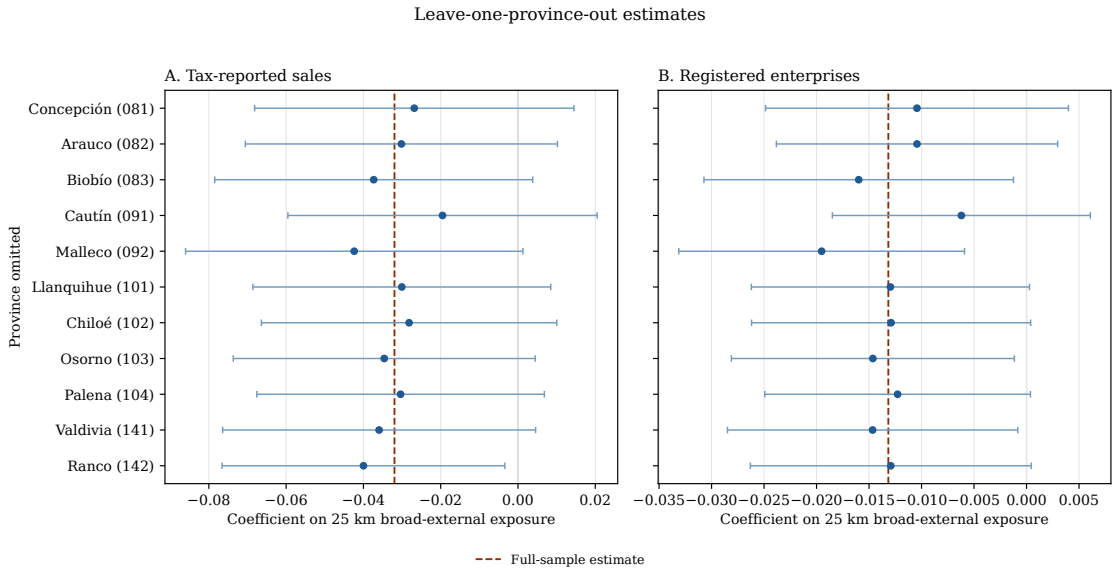


Figure B.1: Leave-one-province-out estimates for formal activity. Each point reports the estimate obtained after omitting the indicated province. Vertical lines show the corresponding full-sample estimates, and horizontal bars report 95 percent confidence intervals.

B.5 Distance sensitivity

The main specification measures external conflict within 25 kilometers of each urban area. To assess whether the formal-activity results depend on this particular spatial window, Table B.5 reconstructs external exposure using cumulative radii from 10 to 60 kilometers. The coefficients remain negative for both registered enterprises and tax-reported sales throughout the range. I interpret this exercise as sensitivity to the spatial window rather than as evidence for an optimal radius of influence.

Table B.5: Cumulative external-conflict windows and formal activity

External window	Registered enterprises	Tax-reported sales
Within 10 km	-0.0133* (0.0078)	-0.0209 (0.0245)
Within 20 km	-0.0138** (0.0055)	-0.0292 (0.0195)
Within 25 km (preferred)	-0.0132** (0.0065)	-0.0320* (0.0189)
Within 30 km	-0.0181*** (0.0064)	-0.0478*** (0.0163)
Within 40 km	-0.0172*** (0.0062)	-0.0456** (0.0225)
Within 50 km	-0.0170** (0.0070)	-0.0414* (0.0238)
Within 60 km	-0.0166** (0.0079)	-0.0451* (0.0246)

Notes: Each row replaces the preferred 25-kilometer exposure measure with log external conflict accumulated within the indicated radius. The event family, within-area conflict, area and region-by-year effects, demographic trends, and area clustering are unchanged. History uses exact-25-km broad-external events in 1990–2004 in every row; only current external exposure varies. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.6 Forestry exclusions

The negative enterprise association persists after removing firms assigned to forestry activities (Table B.6). Narrow forestry excludes forest nurseries and silviculture, logging, nonwood forest collection, and forestry services (SII groups 021–024). Broad forestry also excludes sawmilling, wood and related products, and pulp and paper manufacturing (groups 161, 162, and 170). The table reports both definitions separately and then excludes aquaculture (group 032) alongside each.

Enterprise counts reconcile with the published municipal totals. All five enterprise estimates use 104 areas over 2005–2021. The coefficient changes from −0.0132 for all enterprises to −0.0122 without narrow forestry and −0.0127 without broad forestry. Joint exclusion of aquaculture leaves both coefficients close to their forestry-only values. The 95 percent confidence intervals for the exclusion estimates include zero.

Sales remain negatively associated with nearby conflict after the same exclusions, but their estimates are less precise. Subtracting excluded subsector sales from the official total changes the coefficient from −0.0320 to −0.0219 under narrow forestry and −0.0233 under broad forestry. Summing retained subsectors instead gives −0.0425 and −0.0461, respectively, against −0.0508 for the corresponding all-subsector total. Published subsector sums need not equal the official municipal total. These are classifications of enterprises by assigned activity; the remaining firms can still sell to the forestry sector, and municipal tax addresses do not locate their activity inside urban areas.

B.7 Aquaculture exclusion

Los Lagos contains the largest concentration of aquaculture activity in the study corridor. Because this sector represents a large share of observed sales in the region, a sector-specific downturn could potentially contribute to the aggregate formal-activity relationship. I therefore reconstruct both SII outcomes after excluding subsector 032 (Acuicultura).

Excluding aquaculture leaves all 104 areas and enterprises almost unchanged: −0.013162 becomes −0.013152. Suppression of subsector sales restricts the comparison to 1,066 common area-years. Total sales in that sample gives −0.0124 (SE 0.0216), compared with −0.0144 (SE 0.0216) without aquaculture. Both estimates are imprecise.

B.8 Margins of adjustment

Recorded terminations count formal cessations that remain current at the date of extraction. Taxpayers that subsequently restarted are excluded. This retrospective administrative measure covers the taxpayer universe rather than only the enterprises in the stock series. Both terminations and the SII series published as first registrations enter as $\log(1 + x)$, without reversing their signs. A negative termination coefficient therefore means fewer recorded terminations, not an increase in exits.

The reference termination window is 2005–2015. SII documents administrative actions in 2016

Table B.6: Formal activity excluding forestry and related manufacturing

Outcome definition	Coefficient	SE	p-value	Area-years
Panel A. Registered enterprises				
All activity	-0.0132	0.0065	0.045	1,768
Excluding narrow forestry	-0.0122	0.0066	0.067	1,768
Excluding broad forestry	-0.0127	0.0067	0.062	1,768
Excluding narrow forestry and aquaculture	-0.0122	0.0066	0.068	1,768
Excluding broad forestry and aquaculture	-0.0127	0.0067	0.063	1,768
Panel B. Sales: official total minus excluded sectors				
All activity	-0.0320	0.0189	0.094	1,768
Excluding narrow forestry	-0.0219	0.0194	0.263	1,768
Excluding broad forestry	-0.0233	0.0195	0.235	1,768
Excluding narrow forestry and aquaculture	-0.0226	0.0195	0.249	1,768
Excluding broad forestry and aquaculture	-0.0240	0.0196	0.223	1,768
Panel C. Sales: sum of retained subsectors				
All activity	-0.0508	0.0222	0.024	1,765
Excluding narrow forestry	-0.0425	0.0243	0.083	1,765
Excluding broad forestry	-0.0461	0.0248	0.066	1,765
Excluding narrow forestry and aquaculture	-0.0411	0.0243	0.094	1,765
Excluding broad forestry and aquaculture	-0.0446	0.0247	0.074	1,765

Notes: Narrow forestry excludes SII groups 021–024; broad forestry also excludes 161, 162, and 170. Group 162 includes a mixed category of wood, cork, straw, and plaiting products. Aquaculture is group 032. Enterprise counts sum firms by assigned activity. Sales in Panel B equal the official municipal total in UF less the published sales of excluded subsectors; Panel C sums sales of retained subsectors. For these sales constructions, literal published ∗ cells are set to zero while numeric cells are unchanged. The separate aquaculture check in Table B.7 instead retains only published numeric cells in its paired sales sample. Outcomes are natural logs on positive support; three zero subsector sums leave Panel C with 1,765 area-years. Every regression uses log(1 + contemporary external main conflict within exact 25 km), log(1 + within-area conflict), area and region-by-year fixed effects, and linear trends in log(1 + 2002 municipal population), 2002 rural share, and log(1 + cumulative 1990–2004 external conflict within 25 km). Standard errors are clustered by area; p-values are shown in the table. The sample covers 104 areas in 2005–2021.

Table B.7: Formal activity excluding aquaculture

Outcome	Main estimate	Excluding subsector 032	Area-years
Registered enterprises	-0.0132** (0.0065)	-0.0132** (0.0065)	1,768
Tax-reported sales	-0.0124 (0.0216)	-0.0144 (0.0216)	1,066

Notes: Aquaculture corresponds to SII subsector 032. The enterprise comparison uses the complete 104-area panel. Because subsector-level sales are suppressed in some municipality-years, the sales comparison uses the same observable sample before and after excluding aquaculture. All specifications otherwise reproduce the main area-level model. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and 2017 that increased recorded terminations nationally. Before those actions, the coefficient is −0.0116 (SE 0.0218; $p = 0.595$; 1,144 area-years), with a 95 percent interval of $[-0.0549, 0.0317]$. This estimate provides no clear evidence of either higher or lower recorded terminations. The full 2005–2021 estimate is −0.0319 (SE 0.0160; $p = 0.0497$); excluding only 2016–2017 gives −0.0309 (SE 0.0164; $p = 0.063$). These longer-window estimates are sensitivity checks. Omitting the two intervention years retains observations from both sides of the administrative change and does not establish their comparability. The difference between the estimates does not identify the effect of the administrative intervention.

Administrative registration activity gives −0.0144 (SE 0.0073; $p = 0.053$) over 2005–2021, alongside an enterprise-stock estimate of −0.0132 ($p = 0.045$). On the common 2005–2015 window, registration activity gives 0.0066 (SE 0.0087; $p = 0.447$) and enterprise stock gives −0.0180 (SE 0.0083; $p = 0.032$). The negative registration association therefore does not hold across both windows. The SII archive also publishes identical data rows in files labeled *primera inscripción* and *actividades inscritas*, although SII defines these concepts separately. The analysis reads the former file and interprets it as administrative registration activity, rather than a clean count of firm births.

Table B.8: Enterprise stocks, administrative registration activity, and terminations

Outcome	Window	Coefficient	SE	p	N
Enterprise stock	2005–2021	-0.0132**	0.0065	0.045	1,768
Registration activity	2005–2021	-0.0144*	0.0073	0.053	1,768
Current terminations	2005–2015	-0.0116	0.0218	0.595	1,144
Terminations, full window	2005–2021	-0.0319**	0.0160	0.0497	1,768
Terminations, omit 2016–2017	2005–2021	-0.0309*	0.0164	0.063	1,560

Notes: All rows use the main area-level specification and 104 areas, with area-clustered standard errors. Enterprise stock is measured in logs; registration activity and current terminations use $\log(1 + x)$. The reference termination window is 2005–2015, before SII administrative actions increased terminations in 2016–2017. The last two rows are sensitivity checks; the final row excludes 2016 and 2017. Terminations include zero-count area-years. Current terminations exclude taxpayers that restarted before extraction. Registration activity is the series published as first registrations, subject to the source duplication described above. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Fewer registrations and fewer terminations would move enterprise stock in opposite directions under a clean accounting identity. These administrative series have different coverage from the enterprise stock, and coefficients on their log transformations cannot be subtracted to decompose its change. Restricting the window also leaves the retrospective exclusion of restarters unchanged. The estimates do not identify the contribution of entry and exit to changes in enterprise stocks.

B.9 Transport-related conflict and adjustment margins

The enterprise-stock and registration associations are more negative for conflict at transport locations than for conflict at other locations. I classify attacks, coercive actions, and land invasions coded by MACEDA as ROAD, BRIDGE, or HIGHWAY as transport events; all remaining events in these three families are nontransport events. Each event must occur outside the receiving urban area and within 25 km of its footprint, measured by exact event-point-to-footprint distance. The

log of one plus each annual count enters the preferred area-level specification jointly. Location type is strongly associated with event family, so I also compare transport and nontransport locations within attacks. The transport-minus-nontransport difference in the SII series published as first registrations is nearly unchanged in that comparison, with lower precision. Enterprise stock follows the same negative ordering. The transport–nontransport differences for recorded terminations are negative but imprecisely estimated in both the full window and the pre-2016 sample.

Table B.9: Transport-related conflict and administrative adjustment margins

Outcome / specification	Transport	Nontransport	Transport − nontransport	Equality p-value
Enterprise stock	-0.0266**	-0.0081	-0.0185*	0.077
	(0.0105)	(0.0057)	(0.0104)	
SII first-registration series	-0.0400***	-0.0047	-0.0353**	0.028
	(0.0120)	(0.0076)	(0.0158)	
First-registration series: attacks only	-0.0394***	-0.0039	-0.0355*	0.078
	(0.0134)	(0.0105)	(0.0199)	
First-registration series: episode counts	-0.0341**	-0.0050	-0.0290	0.109
	(0.0133)	(0.0085)	(0.0179)	
First-registration series: attack episode counts	-0.0385**	-0.0049	-0.0336	0.122
	(0.0149)	(0.0109)	(0.0215)	
Terminations, 2005–2015	-0.0803	0.0038	-0.0841	0.160
	(0.0527)	(0.0218)	(0.0595)	
Terminations: full-window sensitivity	-0.0643**	-0.0189	-0.0455	0.283
	(0.0318)	(0.0177)	(0.0421)	

Notes: Transport events are main-family MACEDA events coded ROAD, BRIDGE, or HIGHWAY; nontransport events are the remaining main-family events. External counts use exact distance of at most 25 km from the receiving area and exclude events inside it. The two log(1+count) exposures enter jointly. All models include area and region-by-year fixed effects, three preferred predetermined trends, and contemporaneous log(1+within-area main-family events); the within-area event-count control is retained in episode specifications. Standard errors are clustered by the 104 fixed urban areas. Enterprise stock is in logs; the SII first-registration and termination series are in log(1+x). The 2005–2021 sample has 1,768 area-years; the pre-2016 termination sample has 1,144. Differences are formal linear contrasts, with equality p-values. Episode counts assign transport class if any eligible main-family record in the full episode has a transport location (attacks analogously) and count distinct episodes per receiving area-year without moving their dates. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Counting related MACEDA records as episodes preserves most of the negative registration difference, but the transport–nontransport contrasts are less precise. An episode is classified as transport if any eligible main-family record in the full MACEDA file has a transport location; the attack comparison applies the same rule to attack records. Exposure counts each distinct episode once per receiving area-year when at least one qualifying record lies outside that area and within 25 km. An episode can therefore contribute in more than one year, and timing remains tied to qualifying records rather than the episode root.

Transport and nontransport event–area pairs have similar distances: their mean and median distances are 13.11 and 13.37 km for transport events, against 13.58 and 13.78 km for other events.

Attack-only distances are also similar. With the first-registration series as the outcome, a model entering transport and nontransport conflict exposure jointly in three fixed bands gives negative transport-minus-nontransport differences at 0–10 and > 10–20 km, but not at > 20–25 km. The three equality restrictions reject jointly ($F(3, 103) = 3.557$, $p = 0.0169$). This pattern does not identify an optimal radius or a causal distance threshold.

The registration-source limitation and the pre-2016 reference window for terminations follow Appendix B.8. The full-window termination comparison is retained as a sensitivity check.

C Timing and conflict escalations

C.1 Continuous lead-lag results

The continuous lead–lag projection includes future, contemporaneous, and lagged values of annual external conflict exposure. For presentation, coefficients on future exposure are plotted at negative horizons. These terms capture whether current outcomes covary with exposure realized in subsequent years, conditional on the remaining exposure terms and controls. The contemporaneous and lagged coefficients describe the association with current and past exposure. Because conflict exposure is continuous and recurrent, the coefficients describe temporal patterns in the exposure–outcome relationship rather than event time around a treatment onset.

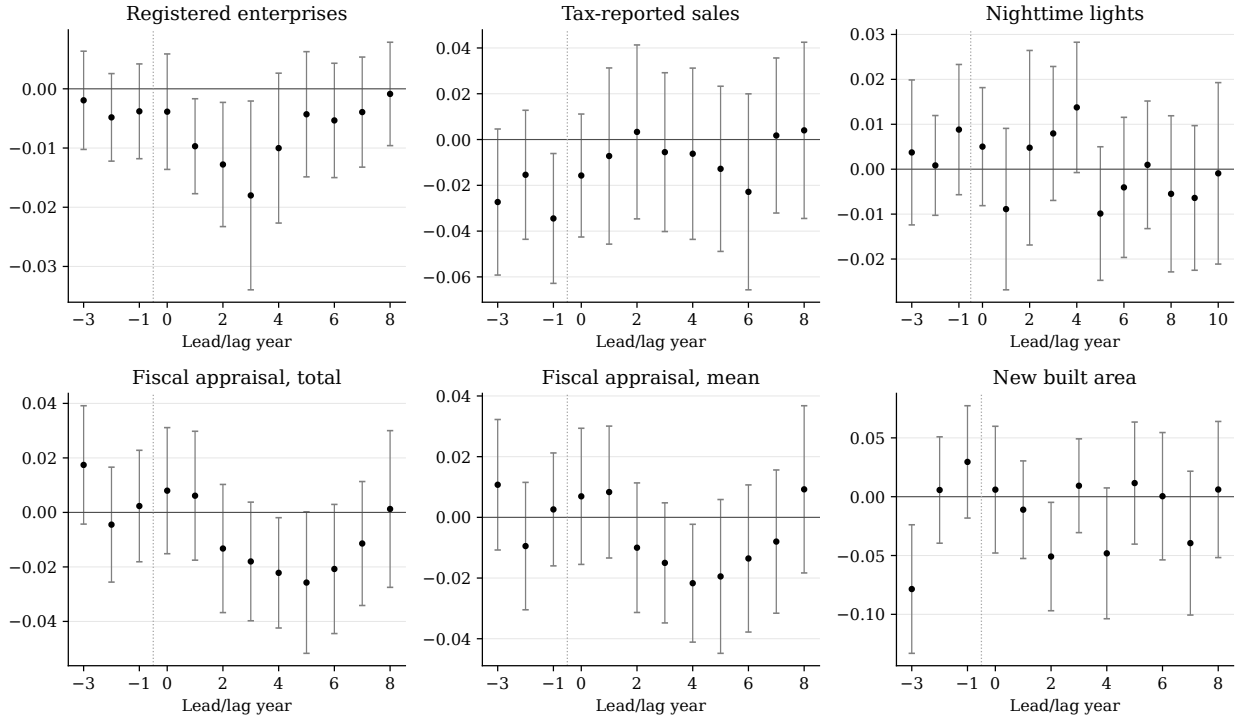


Figure C.1: Continuous lead-lag profiles for urban-area outcomes. Negative horizons denote future exposure terms, zero denotes current exposure, and positive horizons denote lagged exposure. Bars show 95 percent confidence intervals based on area-clustered standard errors.

Table C.1: Summary of continuous lead-lag profiles

	Sales	Enterprises	Nighttime lights	Appraisal total	Appraisal mean	New built area
Joint leads -3/-1	2.026	0.641	0.497	1.286	1.007	3.076**
	p=0.115	p=0.590	p=0.685	p=0.283	p=0.393	p=0.031
Average current/lag 0–2	-0.007	-0.009**	0.000	0.000	0.002	-0.019
	(0.015)	(0.004)	(0.006)	(0.010)	(0.009)	(0.013)
Average lags 3–5	-0.008	-0.011*	0.004	-0.022**	-0.019**	-0.009
	(0.015)	(0.006)	(0.004)	(0.009)	(0.009)	(0.014)
Average lags 6–8	-0.006	-0.003	-0.003	-0.010	-0.004	-0.011
	(0.016)	(0.004)	(0.005)	(0.011)	(0.011)	(0.015)
Observations	1,456	1,456	1,964	936	936	1,872
Areas	104	104	104	104	104	104
Window	2005–2018	2005–2018	2000–2018	2010–2018	2010–2018	1998–2015

Notes: The joint-lead row reports the Wald statistic and its p-value for the three future-exposure coefficients. Current-and-lag rows are covariance-aware averages of the indicated current and lagged coefficients. All models use area and region-by-year effects, within-area conflict, and area-clustered SEs. Formal outcomes and appraisal use all three trends; full-window lights and built-area models omit them. Windows differ with outcome availability and lead/lag support. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The joint future-exposure tests give $p = 0.590$ for enterprises, 0.115 for sales, 0.685 for lights, 0.283 for total appraisal, and 0.393 for mean appraisal. Newly built area shows clearer pre-movement ($p = 0.031$). Its timing profile therefore requires particular caution; absence of a rejection for another outcome does not establish exogeneity.

C.2 Major conflict escalations

The continuous lead–lag results describe recurrent changes in exposure; a separate spike exercise asks whether unusually large annual escalations are followed by a sharper change in urban outcomes. A past-window spike requires at least three external-conflict events and a count exceeding the mean plus four sample standard deviations of the preceding ten years. All ten prior years must be observed. This exercise is secondary to the continuous timing evidence.

I estimate non-normalized effects with the non-absorbing estimator of de Chaisemartin and D'Haultfœuille (2024). Event time starts at the first zero-to-one change; the observed path is retained afterwards. Adjusted models include the three preferred trend regressors and current within-area conflict. Comparisons are restricted to the same region using the estimator's nonparametric regional adjustment rather than TWFE dummy regressors. Characteristics predate these post-2005 panels. Identification still requires conditional parallel trends and no anticipation for the supported switchers and comparison areas.

Formal business outcomes show no clear contraction after a past-window spike (Figure C.2 and Table C.2). At five years, the enterprise estimate is -0.0025 (SE 0.0258); sales are also imprecisely estimated. The five-year enterprise estimate draws on 29 switchers, none of whose supported paths contains another spike by that horizon. Total appraisal is lower at five years, -0.1631 (SE 0.0918), but only 11 appraisal switchers contribute and the joint placebo test gives $p = 0.087$. The delayed appraisal pattern therefore has limited support and cannot be read as a clear response to escalation.

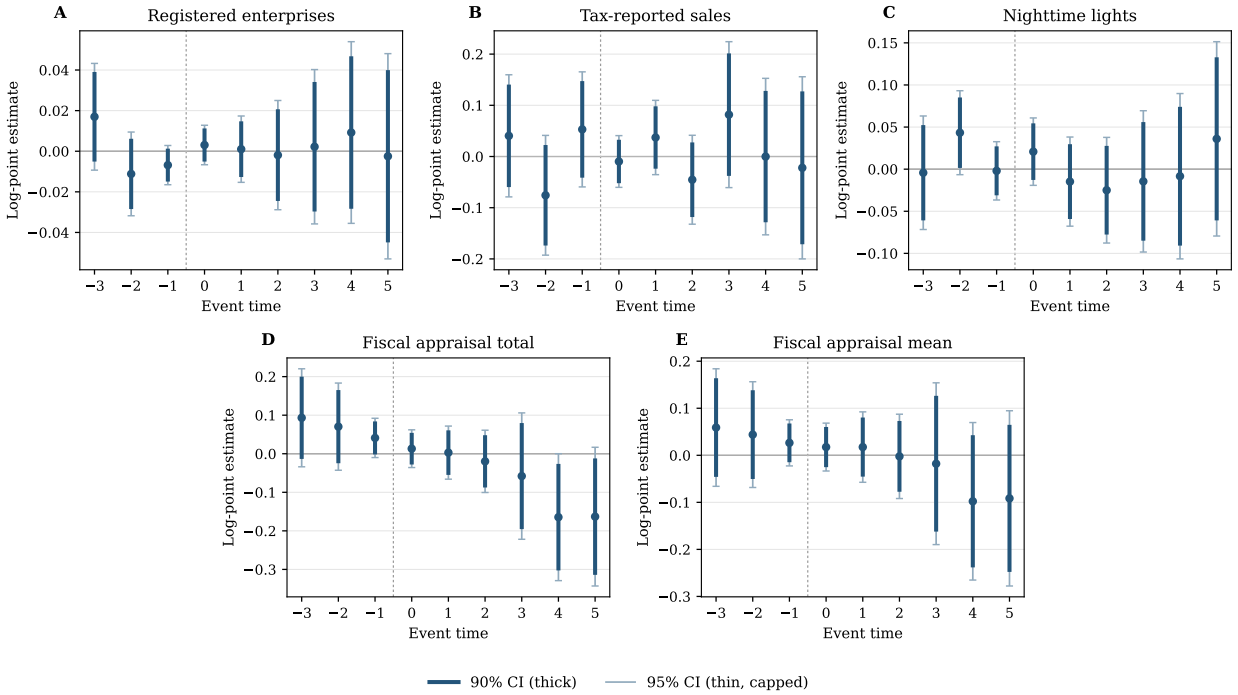


Figure C.2: Past-window spikes and urban-area outcomes. All five panels use the past-window spike definition. Points report non-normalized effects from the de Chaisemartin–D'Haultfoeuille estimator; thick vertical bars show 90 percent confidence intervals and thin capped bars show 95 percent confidence intervals. The annual treatment remains non-absorbing after the first within-panel switch. Estimates at negative event times are placebo estimators, and the dotted vertical line separates them from post-switch effects. All specifications include the three predetermined trends and within-area conflict adjustment, with same-region comparisons as described in the text. Inference uses the estimator's area-level variance.

Table C.2: Dynamic effects following past-window conflict escalations

Outcome	t	$+1$	$+2$	$+3$	$+5$	Placebo p
Registered enterprises	0.003 (0.005)	0.001 (0.008)	-0.002 (0.014)	0.002 (0.019)	-0.002 (0.026)	0.181
Tax-reported sales	-0.010 (0.026)	0.037 (0.037)	-0.045 (0.044)	0.082 (0.073)	-0.022 (0.091)	0.151
Urban nighttime lights	0.021 (0.020)	-0.015 (0.027)	-0.025 (0.032)	-0.015 (0.043)	0.036 (0.059)	0.135
Fiscal appraisal total	0.013 (0.025)	0.003 (0.035)	-0.020 (0.041)	-0.058 (0.084)	-0.163* (0.092)	0.087
Fiscal appraisal mean	0.017 (0.026)	0.017 (0.038)	-0.002 (0.046)	-0.018 (0.088)	-0.092 (0.095)	0.387

Notes: Cells report non-normalized DCDH estimates, with standard errors in parentheses. The final column reports the joint p-value for the three placebo estimators. A past-window spike requires at least three events and a count above the preceding ten-year mean plus four sample standard deviations, with all ten preceding years observed. The annual treatment path remains non-absorbing after the first switch. Adjusted models use the three preferred trends, within-area conflict, same-region comparisons, and area-level inference. Areas must have zero treatment at the initial outcome year. The total-appraisal placebo test gives $p = 0.087$, limiting a causal interpretation. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Within-area spatial incidence

The aggregate estimates describe how formal activity varies across urban areas over time. Table D.1 instead compares fixed 1-kilometer grid cells within the same area-year. At a given radius R, an external event at distance d from a grid-cell centroid contributes $\max\{0, 1 - d/R\}$ to S_R. Thus, an event at the cell receives weight one, an event halfway to the stated radius receives weight one-half, and an event at or beyond the radius receives zero. S_R sums these weighted attacks, coercive actions, and land invasions outside all urban systems, and the regression uses $\log(1 + S_R)$. Each specification also includes direct local main conflict, $\log(1 + L)$, inside the same grid cell.

Grid-cell fixed effects absorb persistent differences across locations, and area-by-year fixed effects absorb shocks shared within an urban area. The selected legal-entity count is more negative at 5 and 10 kilometers, but is close to zero or positive at 20, 25, and 30 kilometers. Commercial-use share is negative at 10, 20, 25, and 30 kilometers, with a similarly negative point estimate at 5 kilometers. Nighttime lights, fiscal appraisal, and new built area do not show a common negative spatial gradient. This local within-area spatial-incidence profile does not identify aggregate urban decline, changes in the location of activity, or a causal radius. Legal-entity locations come from selected headquarters records and cannot be read as all firms or as locations of production or sales.

Table D.1: Within-area spatial incidence across external radii

	Selected legal entities	Urban nighttime lights	Fiscal appraisal total	Fiscal appraisal mean	Commerce-use share	New built area
5 km	-0.1141***	0.0356	0.0497	0.0217	-0.1012*	-0.0010
	(0.0275)	(0.0475)	(0.0902)	(0.0637)	(0.0561)	(0.0282)
10 km	-0.0564**	-0.0057	0.0539	0.0190	-0.1578***	0.0467
	(0.0272)	(0.0582)	(0.0704)	(0.0594)	(0.0522)	(0.0322)
20 km	0.0419	-0.0878	0.1051	0.0112	-0.1677**	0.0154
	(0.0465)	(0.0836)	(0.0979)	(0.0508)	(0.0717)	(0.0574)
25 km (preferred)	0.0428	-0.0659	0.1332	0.0305	-0.1618**	0.0161
	(0.0648)	(0.0805)	(0.0920)	(0.0526)	(0.0629)	(0.0440)
30 km	0.0446	-0.0452	0.1536*	0.0606	-0.1637***	0.0230
	(0.0783)	(0.0779)	(0.0851)	(0.0507)	(0.0614)	(0.0364)
Observations	27,591	27,591	19,014	19,014	15,901	17,853
Fixed 1-km grid cells	1,623	1,623	1,623	1,623	1,623	1,623
Areas	104	104	104	104	104	104
Outcome years	2005–2021	2005–2021	2010–2021	2010–2021	2012–2021	2005–2015

Notes: Each column includes fixed effects for 1-kilometer grid cells and area-by-year fixed effects. For radius R, the R-only external exposure assigns an eligible event at distance d a weight $\max\{0, 1 - d/R\}$: one at the grid cell, one-half halfway to R, and zero at or beyond R. $\log(1 + S_R)$ sums these weighted attacks, coercive actions, and land invasions outside all urban systems. Direct local main conflict, $\log(1 + L)$, inside the same grid cell is included in every specification. Standard errors are clustered by fixed urban area. Coefficients and standard errors are reported to four decimal places. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Municipal-versus-urban interpretation

E.1 Registered legal entities

The SII outcomes used in the main analysis are assigned through municipal tax addresses and therefore do not mechanically locate economic activity inside the fixed urban footprints. I examine this measurement issue using the registered legal-entity (*persona jurídica*) panel. Its total count includes legal entities before temporal assignment and cadastral classification. The urban and rural subsets retain entities whose historical headquarters can be matched with high confidence to cadastral parcels.

The urban-headquarters coefficient is −0.0466 (SE 0.0221); rural headquarters give 0.0073 (SE 0.0269). Zero-count area-years are retained. The uncertain rural result does not establish an urban-versus-rural difference or an exclusively urban association. The samples are selected, and headquarters need not identify production or sales locations.

E.2 Rural production linkages

A second concern is that municipality-level SII outcomes could primarily reflect businesses tied to the surrounding rural economy rather than activity associated with urban areas. I examine this possibility using predetermined sectoral linkages from the Chilean input-output matrices (*Matriz Insumo-Producto*, MIP) published by the Banco Central de Chile.

For each sector, I construct two measures of its direct connection to rural production. The rural input linkage is the share of the sector's domestic intermediate inputs purchased from agriculture, fruit growing, livestock, and forestry. The rural customer linkage is the share of the sector's domestic intermediate sales sold to those same rural activities. Fishing and aquaculture are excluded from the

Table E.1: External conflict and registered legal entities

Outcome	Transformation	Coefficient	Clustered SE	Zero area-years
Registered enterprises	$\log(x)$	-0.0132**	0.0065	0
Total registered legal entities	$\log(x)$	-0.0073	0.0107	0
Cadastre-matched legal entities	$\log(1+x)$	-0.0335	0.0230	109
Cadastre urban legal entities	$\log(1+x)$	-0.0466**	0.0221	152
Cadastre rural legal entities	$\log(1+x)$	0.0073	0.0269	913

Notes: The total legal-entity count precedes temporal assignment, cadastral matching, and classification. The cadastral-matched sample is the parent for the urban and rural rows. Urban and rural subsets use historical headquarters matched to cadastral parcels through administrative address records. These linked subsets are selected, and headquarters locations need not identify production or sales locations. Zero-count area-years are retained. Standard errors are clustered by area. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

rural group. The 2003 MIP is the preferred benchmark because it is the closest pre-period matrix to the 2005–2021 estimation window; the 1996 MIP provides an earlier benchmark. The linkage measures are standardized across the ten comparable non-rural sectors retained in the preferred sector panel.

Let L_s denote either standardized linkage for sector s. I estimate

$$Y_{mst} = \beta\,(D_{mt} \times L_s) + \rho\,(I_{mt} \times L_s) + \alpha_{ms} + \gamma_{st} + \delta_{mt} + \varepsilon_{mst}, \tag{4}$$

where D_{mt} is external conflict exposure and I_{mt} is within-area conflict. Area-by-sector fixed effects absorb persistent differences in local sector composition, sector-by-year fixed effects absorb national sectoral shocks, and area-by-year fixed effects absorb all shocks common to sectors in the same area and year. The coefficient β therefore asks whether the formal-activity association is systematically different in sectors that are more structurally connected to rural production.

Table E.2: External conflict and predetermined rural sectoral linkages

Benchmark	Linkage	Sales interaction	Enterprise interaction
2003	Customer	-0.0168 (0.0279)	0.0046 (0.0050)
2003	Input	-0.0008 (0.0382)	-0.0034 (0.0034)
1996	Customer	-0.0167 (0.0378)	0.0021 (0.0054)
1996	Input	0.0033 (0.0369)	-0.0028 (0.0036)

Notes: Input and customer linkages are constructed from the Banco Central de Chile input-output matrices and standardized across the ten sectors in the preferred sample. The 2003 matrix is the main pre-period benchmark and the 1996 matrix provides an earlier benchmark. Each coefficient is the interaction between log external conflict exposure within 25 kilometers and the indicated standardized sectoral linkage. Specifications also include the corresponding interaction with within-area conflict and absorb area-by-sector, sector-by-year, and area-by-year fixed effects. Standard errors are clustered by fixed urban area. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The interaction estimates are small and do not display a systematic negative gradient for either sales or enterprises. The conclusion is similar using the 2003 input and customer linkages and the earlier 1996 benchmark. The aggregate formal-activity association therefore does not appear to be disproportionately concentrated in sectors with stronger structural links to rural production.

The headquarters records locate selected legal entities inside the urban area, while the input-output results show no stronger association in sectors with greater rural production linkages. Neither exercise locates all SII sales or production.

F Historical heterogeneity

I examine whether the association with contemporary external conflict within the preferred 25 km radius varies with two historical characteristics of the receiving area. Historical territorial exposure is measured by the share of terrestrial land in the external 0–25 km catchment covered by *Títulos de Merced* (TDM). Historical Mapuche presence is measured by its share of the urban population in 1992, reconstructed from individual census records and harmonized to the fixed urban-area geography. The two measures capture related but distinct dimensions of the historical setting and enter jointly in the specifications.

Table F.1 reports the 25 km interactions. The TDM enterprise interaction is −0.0089 (SE 0.0110) and the sales interaction is −0.0007 (SE 0.0210). Both are uncertain, as are the 1992 Mapuche-share interactions. Figure F.1 varies only the moderator catchment through 10, 20, 25, 30, and 50 km. Contemporary conflict exposure and cumulative pre-2005 external conflict remain defined at 25 km. These estimates do not establish a systematic historical gradient.

The preferred TDM total-appraisal interaction is −0.0189 (wild $p = 0.319$); mean appraisal gives −0.0087 ($p = 0.587$). Neither survives multiplicity adjustment. Appraisal Mapuche interactions are positive and uncertain. Appraisal is an assessed stock rather than transaction prices. TDM coverage is concentrated: 43 of 104 areas have no titled land in the external 25 km catchment, and much of the positive variation is in Araucanía.

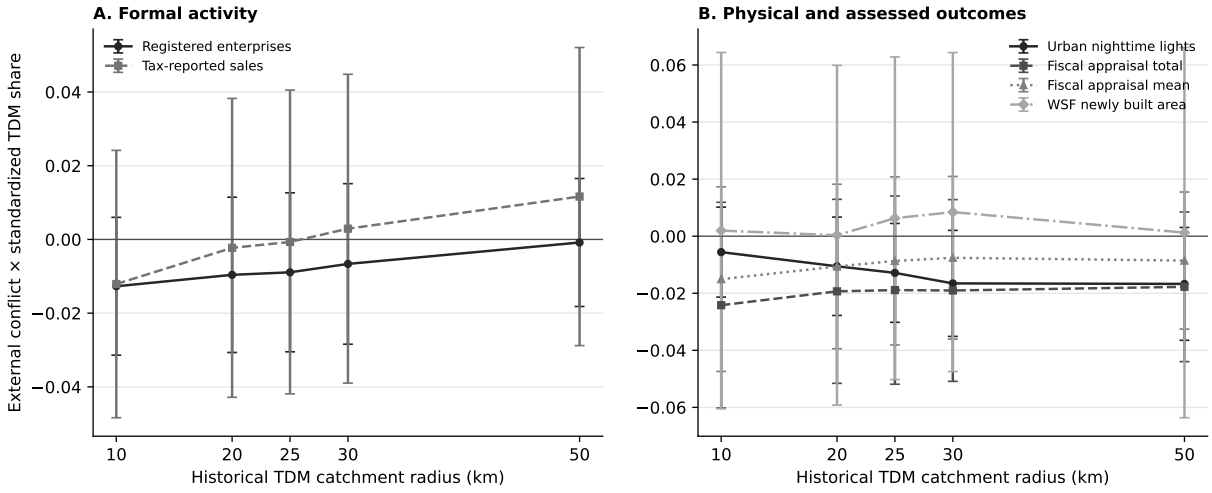
A supplementary specification sets both contemporary external conflict and the TDM catchment to 30 km and allows the external and within-area conflict slopes to vary by region. Across registered enterprises, tax-reported sales, nighttime lights, total and mean fiscal appraisal, and newly built area, it finds no systematic gradient by TDM coverage or 1992 Mapuche share. This differs from Figure F.1, which keeps contemporary conflict at 25 km and varies only the TDM catchment.

Table F.1: Historical territorial exposure, baseline Mapuche presence, and urban-area outcomes

Outcome	TDM share, external 0–25 km β (SE)	Wild p	BH q	1992 Mapuche urban-population share β (SE)	Wild p	BH q
Panel A. Formal activity						
Registered enterprises	-.009 (.011)	.512	.902	.010 (.015)	.663	.909
Tax-reported sales	-.001 (.021)	.975	.991	-.004 (.033)	.918	.975
Panel B. Physical and assessed outcomes						
Urban nighttime lights	-.013 (.009)	.174	.902	.007 (.015)	.682	.909
Fiscal appraisal total	-.019 (.017)	.319	.902	.036 (.024)	.223	.909
Fiscal appraisal mean	-.009 (.015)	.587	.902	.019 (.022)	.469	.909
WSF newly built area	.006 (.029)	.866	.991	-.033 (.038)	.460	.909

Notes: The table reports interactions between external conflict exposure and two standardized historical moderators, estimated jointly. TDM exposure is the share of terrestrial land in the receiving area's external 0–25 km catchment covered by historical *Títulos de Merced*. Mapuche population share is reconstructed from individual records of the 1992 Census and measures the Mapuche share of the historical urban population harmonized to the fixed urban-area geography. The joint sample contains 102 areas; Curaco de Vélez and Palena have no urban population records in the 1992 census geography. All models include area and region-by-year effects, within-area conflict, and moderator-specific trends. Post-2005 panels use the three preferred trends; the longer built-area panel omits them. Lights here use the post-2005 historical-moderator sample. Models also include the corresponding moderator interactions. Standard errors are clustered by fixed urban area and wild-cluster p-values are reported. TDM q-values adjust the 30 radius-by-outcome tests across the five stated catchment boundaries; Mapuche q-values use the pre-specified Mapuche interaction family. Stars use the reported wild-cluster p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure F.1: Historical TDM exposure across alternative catchment boundaries



Notes: Points report interactions between external conflict exposure and standardized TDM land share; bars show 95 percent confidence intervals based on area-clustered standard errors. The 10-, 20-, 25-, 30-, and 50-km catchments are alternative boundaries for the same measure of surrounding historical territorial exposure and are highly correlated. Contemporary conflict exposure uses the preferred 25-km definition in every regression; only the historical moderator catchment varies.

G Institutional response: CONADI

CONADI records provide evidence on formal land-acquisition activity in the same territorial setting. Article 20(a) and 20(b) acquisitions are aggregated to area-years using the official municipality codes and the municipality–urban-area crosswalk. The source reports an acquisition year, although the precise administrative stage represented by that date is not documented. The two outcomes are the log of one plus the number of recorded acquisitions and the log of one plus reported acquired hectares.

Greater external conflict exposure is followed by more recorded CONADI acquisition activity at the two-year lag; the one-year coefficients are imprecisely estimated. The cumulative current-plus-two-lag association is positive for both acquisitions and acquired hectares. This documents continued formal land-acquisition activity in conflict-exposed territories; it does not identify the effect of CONADI acquisitions on conflict or local economic activity.

Table G.1: External conflict exposure and recorded CONADI land acquisition

Outcome	Static	D_t	D_{t-1}	D_{t-2}	Joint p	Cumulative
Acquisition events	0.0751** (0.0320)	0.0675* (0.0343)	-0.0051 (0.0295)	0.0896*** (0.0263)	0.003	0.1519*** (0.0442)
Reported hectares	0.1707* (0.0873)	0.1582* (0.0935)	-0.0525 (0.0923)	0.2456*** (0.0871)	0.022	0.3513** (0.1378)

Notes: The static model uses contemporaneous external conflict exposure. The distributed-lag model includes exposure at t, $t-1$, and $t-2$ jointly. The cumulative column reports the sum of these three coefficients, with its covariance-adjusted standard error. All specifications use the preferred 104-area, 2005–2021 specification and area-clustered standard errors. CONADI outcomes are transformed as $\log(1+x)$. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.